

Database Analysis Report

Database: sales_agent.db

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Team: TNMT

Executive Summary

This PDF contains an automated analysis of the supplied SQLite database. It includes schema exploration, data quality checks, statistical summaries, and visualizations.

Table: dbo_D_Customer

Rows	2632			
Columns	73			
Column	Type	Non-null	Nulls	Unique
Customer Key	int64	2632	0	2632
Company Code	object	2631	1	3
Source Code	object	2631	1	1
Customer Number	float64	2631	1	2631
Bill-to Customer Key	int64	2632	0	1
Customer Category Hrchy Key	int64	2632	0	19
Customer Geography Hrchy Key	int64	2632	0	99
Default Customer Ship-to Key	int64	2632	0	2632
Sales Organization Key	int64	2632	0	21
Customer Name	object	2631	1	2631

Insights

- In table 'dbo_D_Customer', columns ['Payment Terms Code', 'Industry Desc', 'Responsibility Name', 'Sales Territory Code', 'Sales Territory', 'Payment Terms', 'Payment Terms Group', 'Price Group Code', 'Price Group', 'Discount Group Code', 'Discount Group', 'Business Posting Code', 'Business Posting Desc', 'Customer Posting Desc', 'Credit Status', 'Attribute 1 Code', 'Attribute 1 Name', 'Attribute 2 Code', 'Attribute 2 Name', 'Attribute 3 Code', 'Attribute 3 Name', 'Attribute 4 Code', 'Attribute 4 Name', 'Attribute 5 Code', 'Attribute 5 Name', 'Recency Band', 'Recency Band Sort', 'Frequency Band Sort', 'Responsibility Code', 'Industry Code', 'Bill-to Customer Number', 'Customer Status', 'Origin Date', 'Customer Address 2', 'Customer Address 3', 'Frequency Band', 'Customer Postal Code'] have high missingness (>30%). Consider data collection or imputation.
- Column 'Source Code' in 'dbo_D_Customer' has only one unique value (['Demo']).
- Column 'Bill-to Customer Key' in 'dbo_D_Customer' has only one unique value ([-1]).
- Column 'Customer Status' in 'dbo_D_Customer' is entirely empty.
- Column 'Origin Date' in 'dbo_D_Customer' is entirely empty.
- Column 'Customer Address 2' in 'dbo_D_Customer' is entirely empty.
- Column 'Customer Address 3' in 'dbo_D_Customer' is entirely empty.
- Column 'Bill-to Customer Number' in 'dbo_D_Customer' is entirely empty.
- Column 'Customer Type Code' in 'dbo_D_Customer' has only one unique value (['B']).
- Column 'Customer Type Desc' in 'dbo_D_Customer' has only one unique value (['Business']).
- Column 'Industry Code' in 'dbo_D_Customer' is entirely empty.
- Column 'Industry Desc' in 'dbo_D_Customer' is entirely empty.
- Column 'Responsibility Code' in 'dbo_D_Customer' is entirely empty.
- Column 'Responsibility Name' in 'dbo_D_Customer' is entirely empty.
- Column 'Sales Territory Code' in 'dbo_D_Customer' is entirely empty.
- Column 'Sales Territory' in 'dbo_D_Customer' is entirely empty.
- Column 'Payment Terms Code' in 'dbo_D_Customer' is entirely empty.
- Column 'Payment Terms' in 'dbo_D_Customer' is entirely empty.
- Column 'Payment Terms Group' in 'dbo_D_Customer' is entirely empty.
- Column 'Price Group Code' in 'dbo_D_Customer' is entirely empty.
- Column 'Price Group' in 'dbo_D_Customer' is entirely empty.
- Column 'Discount Group Code' in 'dbo_D_Customer' is entirely empty.
- Column 'Discount Group' in 'dbo_D_Customer' is entirely empty.
- Column 'Business Posting Code' in 'dbo_D_Customer' is entirely empty.
- Column 'Business Posting Desc' in 'dbo_D_Customer' is entirely empty.
- Column 'Customer Posting Desc' in 'dbo_D_Customer' is entirely empty.
- Column 'Credit Status' in 'dbo_D_Customer' is entirely empty.
- Column 'Attribute 1 Code' in 'dbo_D_Customer' is entirely empty.
- Column 'Attribute 1 Name' in 'dbo_D_Customer' is entirely empty.
- Column 'Attribute 2 Code' in 'dbo_D_Customer' is entirely empty.

- Column 'Attribute 2 Name' in 'dbo_D_Customer' is entirely empty.
- Column 'Attribute 3 Code' in 'dbo_D_Customer' is entirely empty.
- Column 'Attribute 3 Name' in 'dbo_D_Customer' is entirely empty.
- Column 'Attribute 4 Code' in 'dbo_D_Customer' is entirely empty.
- Column 'Attribute 4 Name' in 'dbo_D_Customer' is entirely empty.
- Column 'Attribute 5 Code' in 'dbo_D_Customer' is entirely empty.
- Column 'Attribute 5 Name' in 'dbo_D_Customer' is entirely empty.
- Column 'Recency Band' in 'dbo_D_Customer' is entirely empty.
- Column 'Recency Band Sort' in 'dbo_D_Customer' is entirely empty.
- Column 'Frequency Band' in 'dbo_D_Customer' is entirely empty.
- Column 'Frequency Band Sort' in 'dbo_D_Customer' is entirely empty.
- Column 'Sales Activity Flag' in 'dbo_D_Customer' has only one unique value ([1]).
- Column 'Deleted Flag' in 'dbo_D_Customer' has only one unique value ([0]).
- Numeric column 'Credit Limit Amount' in 'dbo_D_Customer' has 929 potential outliers (IQR method).
- Numeric column 'Year Acquired' in 'dbo_D_Customer' has 1 potential outliers (IQR method).
- Numeric column 'High Volume Customer Count' in 'dbo_D_Customer' has 86 potential outliers (IQR method).
- Numeric column 'Monetary Band Sort' in 'dbo_D_Customer' has 143 potential outliers (IQR method).
- Numeric column 'Relationship Length Band Sort' in 'dbo_D_Customer' has 653 potential outliers (IQR method).
- Numeric column 'Source Log Key' in 'dbo_D_Customer' has 1 potential outliers (IQR method).
- Strong correlations detected in 'dbo_D_Customer': Customer Key-Default Customer Ship-to Key (1.00), Customer Number-Salesperson Code (1.00), Customer Number-Customer Loyalty Rule Key (1.00), Customer Category Hrchy Code-Market Code (1.00), Salesperson Code-Customer Loyalty Rule Key (1.00), Year Acquired-Source Log Key (1.00). Consider multicollinearity handling.
- Numeric column 'Year Acquired' in 'dbo_D_Customer' is highly skewed (skew=-49.79). Consider transformation.
- Numeric column 'High Volume Customer Count' in 'dbo_D_Customer' is highly skewed (skew=5.26). Consider transformation.
- Numeric column 'Source Log Key' in 'dbo_D_Customer' is highly skewed (skew=51.30). Consider transformation.

Chart: chart1.png

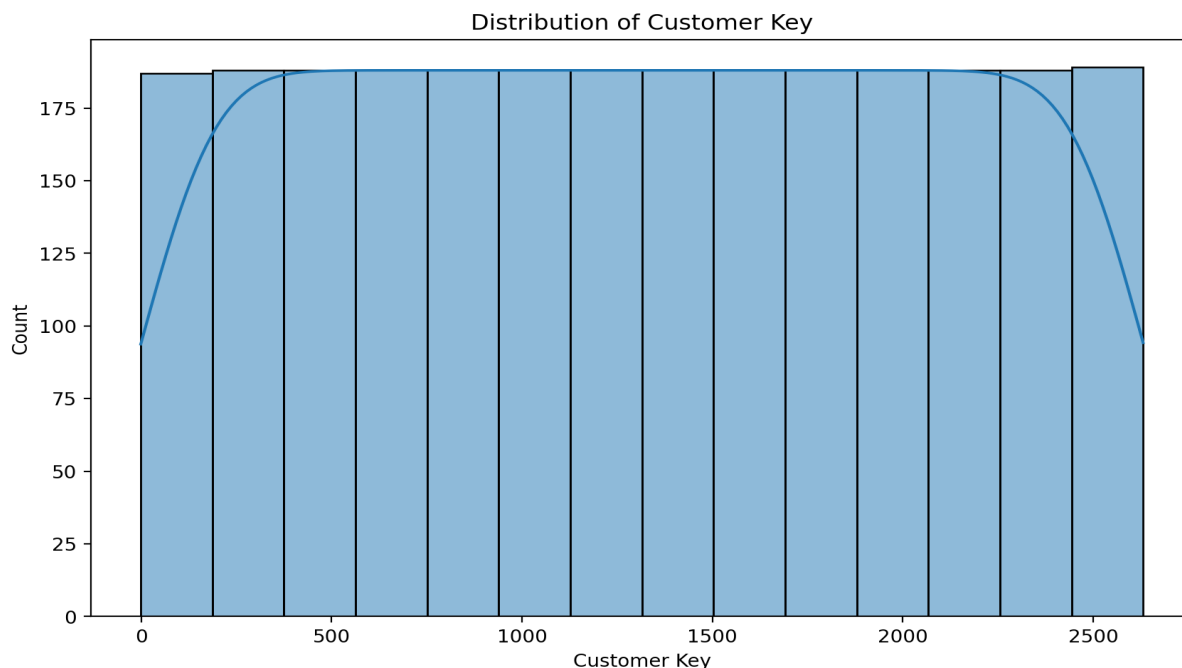


Chart: chart2.png

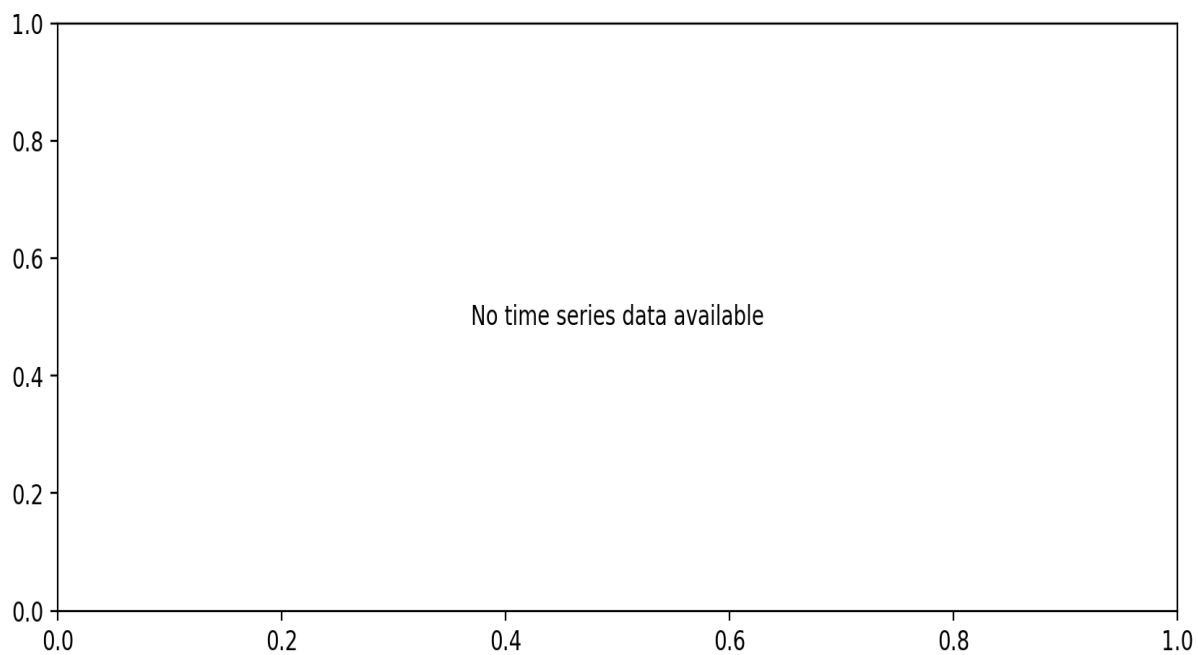


Table: dbo_D_Customer_Category_Hierarchy

Rows	19				
Columns	16				
Column	Type	Non-null	Nulls	Unique	
Customer Category Hrchy Key	int64	19	0	19	
Company Code	object	18	1	3	
Source Code	object	18	1	1	
Customer Category Hrchy Code	float64	18	1	6	
Customer Category Hrchy L1 Code	float64	18	1	6	
Customer Category Hrchy L1 Name	object	18	1	6	
Customer Category Hrchy L2 Code	object	0	19	0	
Customer Category Hrchy L2 Name	object	0	19	0	
Customer Category Hrchy L3 Code	object	0	19	0	
Customer Category Hrchy L3 Name	object	0	19	0	

Insights

- In table 'dbo_D_Customer_Category_Hierarchy', columns ['Customer Category Hrchy L2 Code', 'Customer Category Hrchy L2 Name', 'Customer Category Hrchy L3 Code', 'Customer Category Hrchy L3 Name', 'Customer Category Hrchy L4 Code', 'Customer Category Hrchy L4 Name', 'Customer Category Hrchy L5 Code', 'Customer Category Hrchy L5 Name'] have high missingness (>30%). Consider data collection or imputation.
- Column 'Source Code' in 'dbo_D_Customer_Category_Hierarchy' has only one unique value (['Demo']).
- Column 'Customer Category Hrchy L2 Code' in 'dbo_D_Customer_Category_Hierarchy' is entirely empty.
- Column 'Customer Category Hrchy L2 Name' in 'dbo_D_Customer_Category_Hierarchy' is entirely empty.
- Column 'Customer Category Hrchy L3 Code' in 'dbo_D_Customer_Category_Hierarchy' is entirely empty.
- Column 'Customer Category Hrchy L3 Name' in 'dbo_D_Customer_Category_Hierarchy' is entirely empty.
- Column 'Customer Category Hrchy L4 Code' in 'dbo_D_Customer_Category_Hierarchy' is entirely empty.
- Column 'Customer Category Hrchy L4 Name' in 'dbo_D_Customer_Category_Hierarchy' is entirely empty.
- Column 'Customer Category Hrchy L5 Code' in 'dbo_D_Customer_Category_Hierarchy' is entirely empty.
- Column 'Customer Category Hrchy L5 Name' in 'dbo_D_Customer_Category_Hierarchy' is entirely empty.
- Column 'Deleted Flag' in 'dbo_D_Customer_Category_Hierarchy' has only one unique value ([0]).
- Column 'Source Log Key' in 'dbo_D_Customer_Category_Hierarchy' has only one unique value ([0]).
- Strong correlations detected in 'dbo_D_Customer_Category_Hierarchy': Customer Category Hrchy Code-Customer Category Hrchy L1 Code (1.00). Consider multicollinearity handling.

Chart: [chart1.png](#)

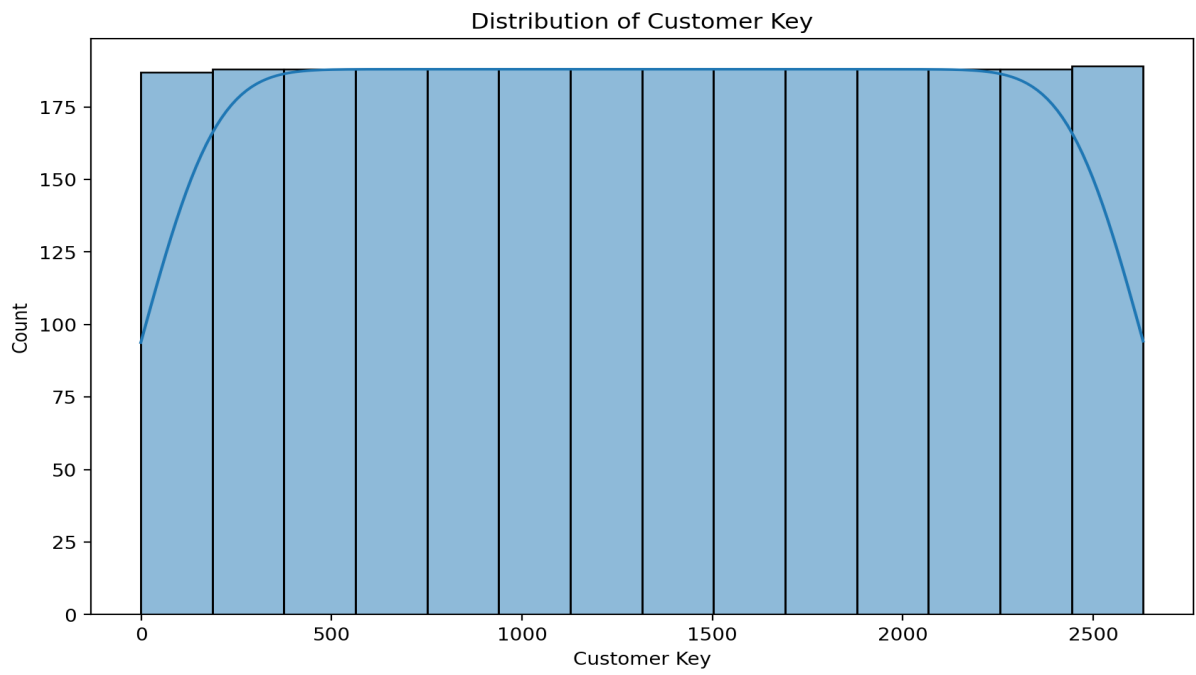


Chart: chart2.png

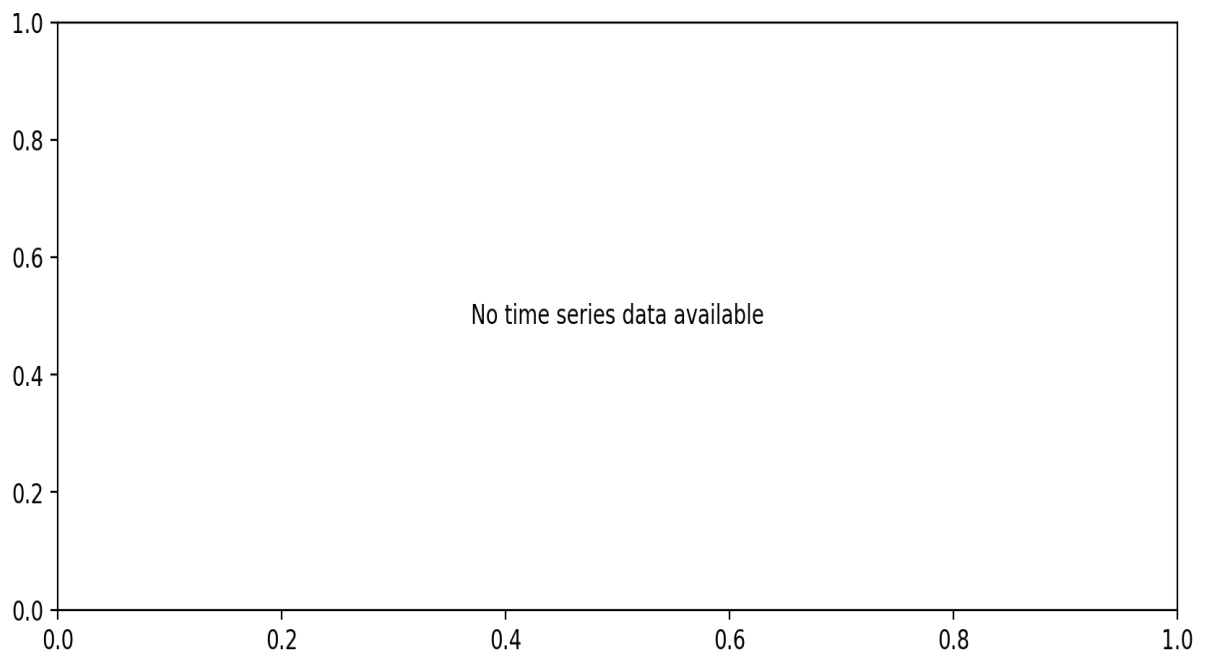


Table: dbo_D_Customer_Geography_Hierarchy

Rows	99				
Columns	16				
Column	Type	Non-null	Nulls	Unique	
Customer Geography Hrchy Key	int64	99	0	99	
Company Code	object	98	1	3	
Source Code	object	98	1	1	
Customer Geography Hrchy Code	object	98	1	98	
Customer Geography Hrchy L1 Code	object	98	1	3	
Customer Geography Hrchy L1 Name	object	98	1	3	
Customer Geography Hrchy L2 Code	object	98	1	13	
Customer Geography Hrchy L2 Name	object	98	1	13	
Customer Geography Hrchy L3 Code	object	98	1	98	
Customer Geography Hrchy L3 Name	object	98	1	98	

Insights

- In table 'dbo_D_Customer_Geography_Hierarchy', columns ['Customer Geography Hrchy L4 Code', 'Customer Geography Hrchy L4 Name', 'Customer Geography Hrchy L5 Code', 'Customer Geography Hrchy L5 Name'] have high missingness (>30%). Consider data collection or imputation.
- Column 'Source Code' in 'dbo_D_Customer_Geography_Hierarchy' has only one unique value (['Demo']).
- Column 'Customer Geography Hrchy L4 Code' in 'dbo_D_Customer_Geography_Hierarchy' is entirely empty.
- Column 'Customer Geography Hrchy L4 Name' in 'dbo_D_Customer_Geography_Hierarchy' is entirely empty.
- Column 'Customer Geography Hrchy L5 Code' in 'dbo_D_Customer_Geography_Hierarchy' is entirely empty.
- Column 'Customer Geography Hrchy L5 Name' in 'dbo_D_Customer_Geography_Hierarchy' is entirely empty.
- Column 'Deleted Flag' in 'dbo_D_Customer_Geography_Hierarchy' has only one unique value ([0]).
- Column 'Source Log Key' in 'dbo_D_Customer_Geography_Hierarchy' has only one unique value ([0]).

Chart: chart1.png

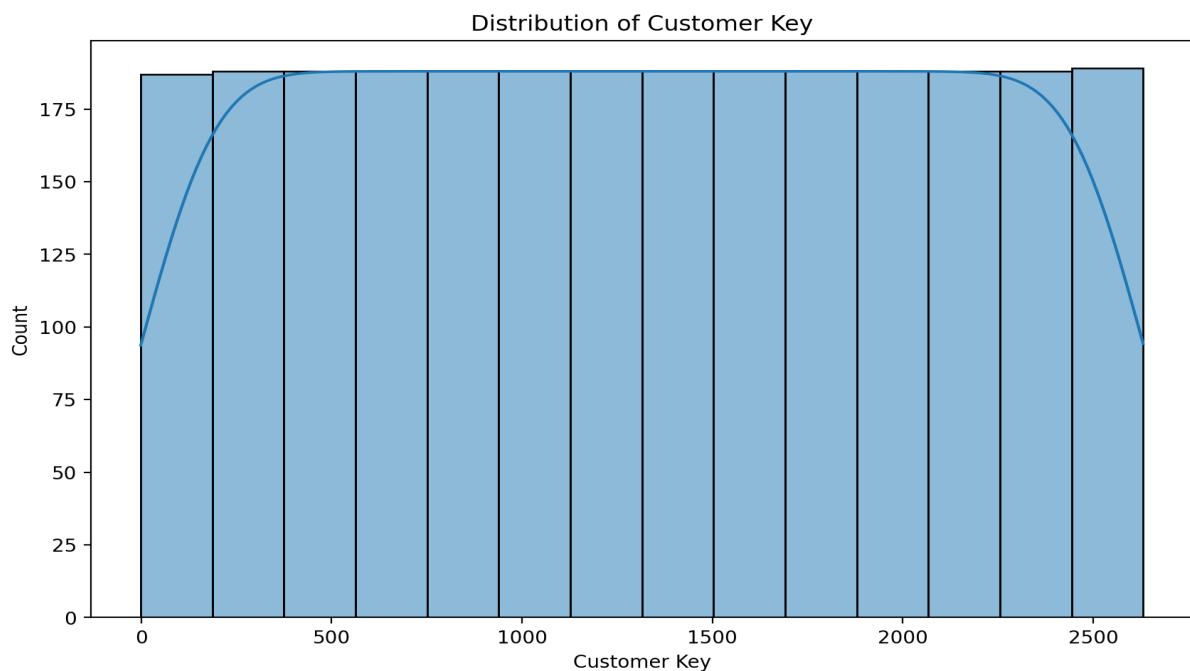


Chart: chart2.png

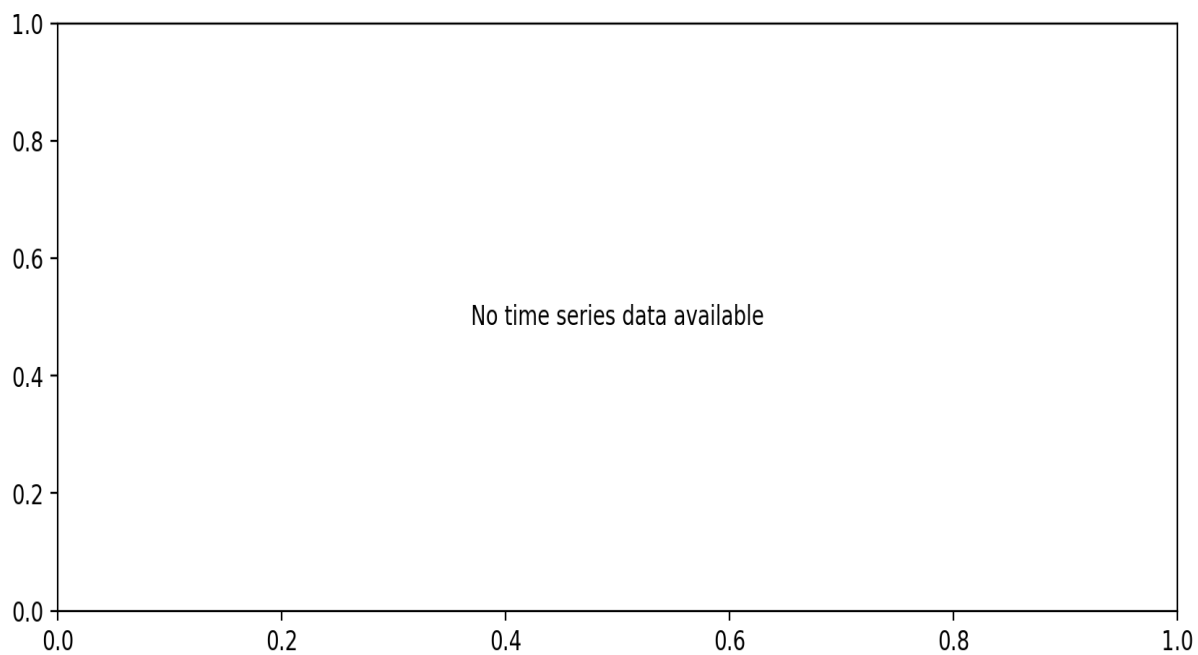


Table: dbo_D_Item

Rows	463			
Columns	62			
Column	Type	Non-null	Nulls	Unique
Item Key	int64	463	0	463
Company Code	object	462	1	3
Source Code	object	462	1	1
Item Type	object	462	1	1
Item Number	object	462	1	154
Buyer Key	int64	463	0	1
Item Category Hrchy Key	int64	463	0	55
Manufacturer Key	int64	463	0	1
Purchaser Key	int64	463	0	1
Vendor Key	int64	463	0	1

Insights

- In table 'dbo_D_Item', columns ['Discount Desc', 'Inventory Posting Code', 'Inventory Posting Desc', 'Discount Code', 'Buyer Code', 'Purchaser Code', 'Manufacturer Number', 'Vendor Item Number', 'Vendor Number', 'Purch Unit of Measure', 'Sales Unit of Measure', 'Attribute 4 Name', 'Attribute 4 Code', 'Item Type Desc', 'Attribute 5 Code', 'Item Status', 'Replenishment Method', 'Attribute 5 Name', 'Attribute 1 Code', 'ABC Code', 'Last Unit Cost', 'Standard Unit Cost', 'Reorder Point', 'Reorder Quantity', 'Safety Stock Quantity', 'Purch Lead Time', 'Mfg Lead Time', 'Attribute 1 Name', 'Attribute 2 Code', 'Attribute 2 Name'] have high missingness (>30%). Consider data collection or imputation.
- Column 'Source Code' in 'dbo_D_Item' has only one unique value (['Demo']).
- Column 'Item Type' in 'dbo_D_Item' has only one unique value (['Item']).
- Column 'Buyer Key' in 'dbo_D_Item' has only one unique value ([-1]).
- Column 'Manufacturer Key' in 'dbo_D_Item' has only one unique value ([-1]).
- Column 'Purchaser Key' in 'dbo_D_Item' has only one unique value ([-1]).
- Column 'Vendor Key' in 'dbo_D_Item' has only one unique value ([-1]).
- Column 'Item Status' in 'dbo_D_Item' is entirely empty.
- Column 'Item Type Code' in 'dbo_D_Item' has only one unique value (['FG']).
- Column 'Item Type Desc' in 'dbo_D_Item' is entirely empty.
- Column 'Vendor Number' in 'dbo_D_Item' is entirely empty.
- Column 'Vendor Item Number' in 'dbo_D_Item' is entirely empty.
- Column 'Manufacturer Number' in 'dbo_D_Item' is entirely empty.
- Column 'Purchaser Code' in 'dbo_D_Item' is entirely empty.
- Column 'Buyer Code' in 'dbo_D_Item' is entirely empty.
- Column 'ABC Code' in 'dbo_D_Item' is entirely empty.
- Column 'Inventory Posting Code' in 'dbo_D_Item' is entirely empty.
- Column 'Inventory Posting Desc' in 'dbo_D_Item' is entirely empty.
- Column 'Discount Code' in 'dbo_D_Item' is entirely empty.
- Column 'Discount Desc' in 'dbo_D_Item' is entirely empty.
- Column 'Unit of Measure' in 'dbo_D_Item' has only one unique value (['EA']).
- Column 'Purch Unit of Measure' in 'dbo_D_Item' is entirely empty.
- Column 'Sales Unit of Measure' in 'dbo_D_Item' is entirely empty.
- Column 'Standard Unit Cost' in 'dbo_D_Item' has only one unique value ([0.0]).
- Column 'Last Unit Cost' in 'dbo_D_Item' has only one unique value ([0.0]).
- Column 'Replenishment Method' in 'dbo_D_Item' is entirely empty.
- Column 'Reorder Point' in 'dbo_D_Item' has only one unique value ([0.0]).
- Column 'Reorder Quantity' in 'dbo_D_Item' has only one unique value ([0.0]).
- Column 'Safety Stock Quantity' in 'dbo_D_Item' has only one unique value ([0.0]).
- Column 'Purch Lead Time' in 'dbo_D_Item' has only one unique value ([0.0]).
- Column 'Mfg Lead Time' in 'dbo_D_Item' has only one unique value ([0.0]).
- Column 'Attribute 1 Code' in 'dbo_D_Item' is entirely empty.

- Column 'Attribute 4 Code' in 'dbo_D_Item' is entirely empty.
- Column 'Attribute 4 Name' in 'dbo_D_Item' is entirely empty.
- Column 'Attribute 5 Code' in 'dbo_D_Item' is entirely empty.
- Column 'Attribute 5 Name' in 'dbo_D_Item' is entirely empty.
- Column 'Sales Activity Flag' in 'dbo_D_Item' has only one unique value ([1]).
- Column 'Deleted Flag' in 'dbo_D_Item' has only one unique value ([0]).
- Column 'Source Log Key' in 'dbo_D_Item' has only one unique value ([0]).
- Numeric column 'Unit Cost' in 'dbo_D_Item' has 30 potential outliers (IQR method).
- Numeric column 'Gross Weight' in 'dbo_D_Item' has 9 potential outliers (IQR method).
- Numeric column 'New Item Count' in 'dbo_D_Item' has 75 potential outliers (IQR method).
- Numeric column 'Purch Activity Flag' in 'dbo_D_Item' has 1 potential outliers (IQR method).
- Numeric column 'Inv Activity Flag' in 'dbo_D_Item' has 1 potential outliers (IQR method).
- Strong correlations detected in 'dbo_D_Item': Unit Price-Unit Cost (0.99), Purch Activity Flag-Inv Activity Flag (1.00). Consider multicollinearity handling.
- Numeric column 'Purch Activity Flag' in 'dbo_D_Item' is highly skewed (skew=21.52). Consider transformation.
- Numeric column 'Inv Activity Flag' in 'dbo_D_Item' is highly skewed (skew=21.52). Consider transformation.

Chart: chart1.png

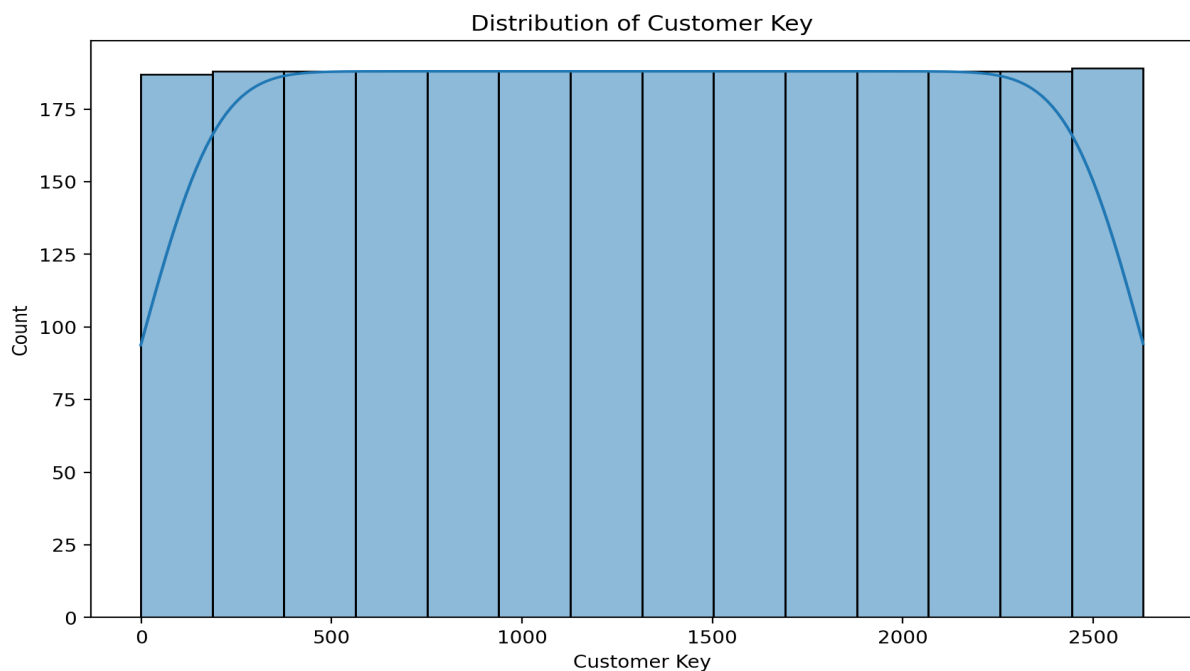


Chart: chart2.png

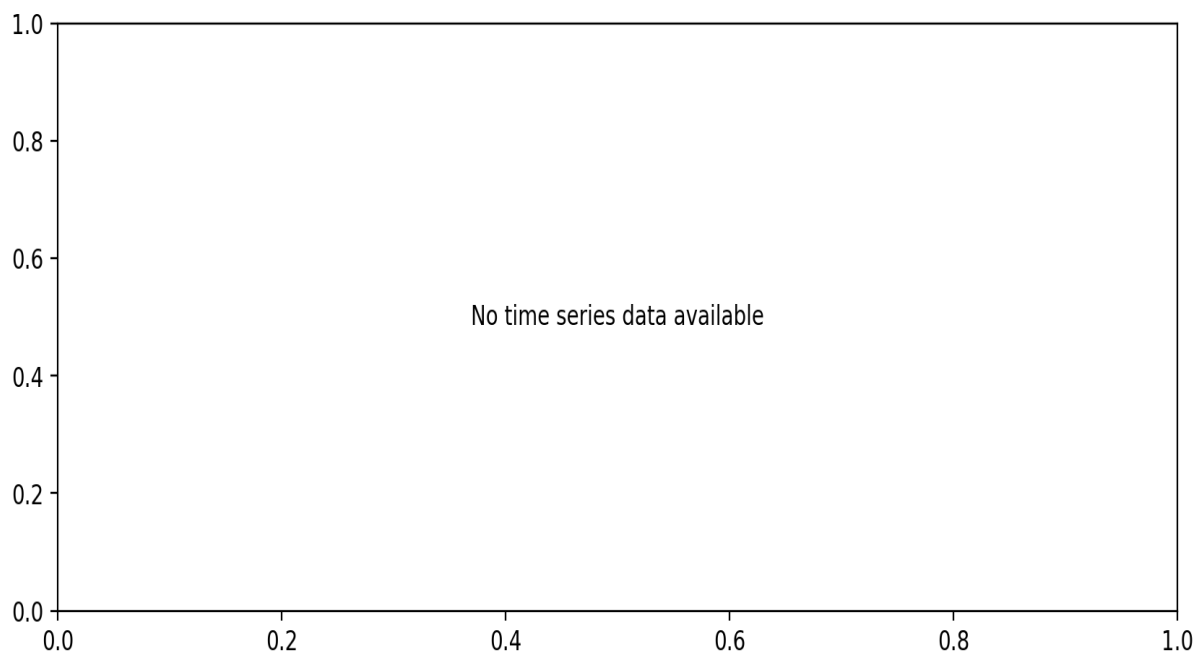


Table: dbo_D_Sales_Organization

Rows	21			
Columns	14			
Column	Type	Non-null	Nulls	Unique
Sales Organization Key	int64	21	0	21
Company Code	object	20	1	3
Source Code	object	20	1	1
Salesperson Code	float64	20	1	20
Salesperson Name	object	20	1	20
Sales Org Hrchy L1 Name	object	20	1	3
Sales Org Hrchy L2 Name	object	20	1	15
Sales Org Hrchy L3 Name	object	0	21	0
Email Address	object	0	21	0
Sales Manager Code	object	0	21	0

Insights

- In table 'dbo_D_Sales_Organization', columns ['Sales Org Hrchy L3 Name', 'Email Address', 'Sales Manager Code', 'Sales Manager Name'] have high missingness (>30%). Consider data collection or imputation.
- Column 'Source Code' in 'dbo_D_Sales_Organization' has only one unique value (['Demo']).
- Column 'Sales Org Hrchy L3 Name' in 'dbo_D_Sales_Organization' is entirely empty.
- Column 'Email Address' in 'dbo_D_Sales_Organization' is entirely empty.
- Column 'Sales Manager Code' in 'dbo_D_Sales_Organization' is entirely empty.
- Column 'Sales Manager Name' in 'dbo_D_Sales_Organization' is entirely empty.
- Column 'Sales Activity Flag' in 'dbo_D_Sales_Organization' has only one unique value ([1]).
- Column 'Deleted Flag' in 'dbo_D_Sales_Organization' has only one unique value ([0]).
- Column 'Source Log Key' in 'dbo_D_Sales_Organization' has only one unique value ([0]).

Chart: chart1.png

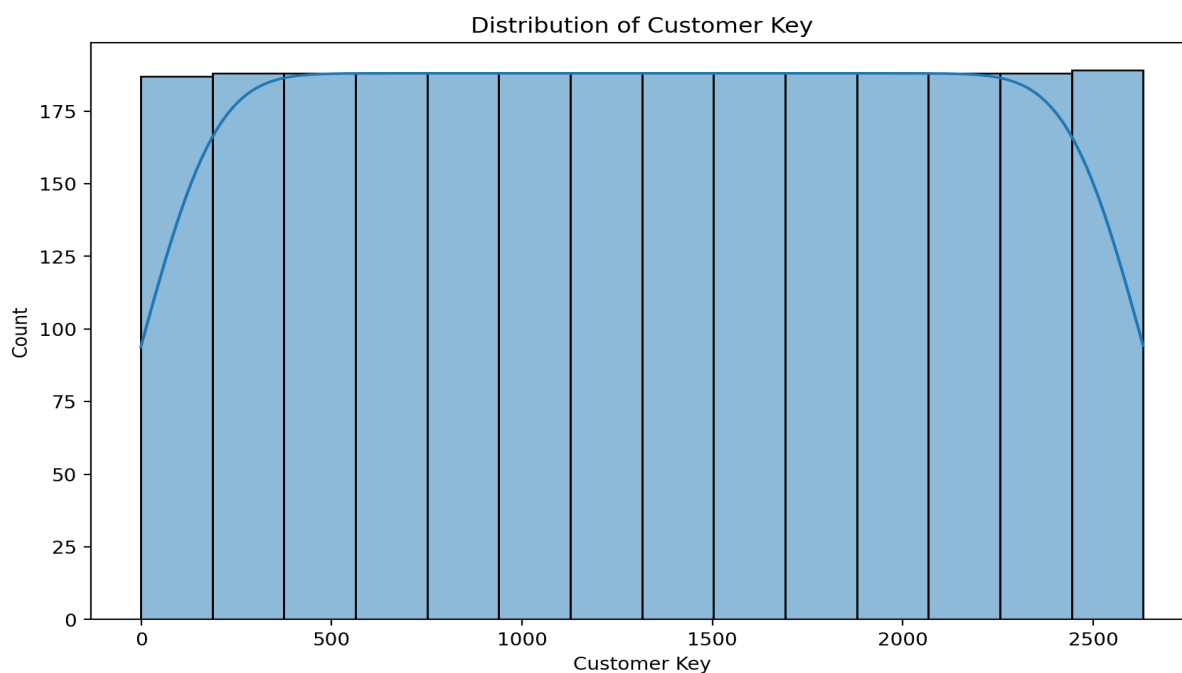


Chart: chart2.png

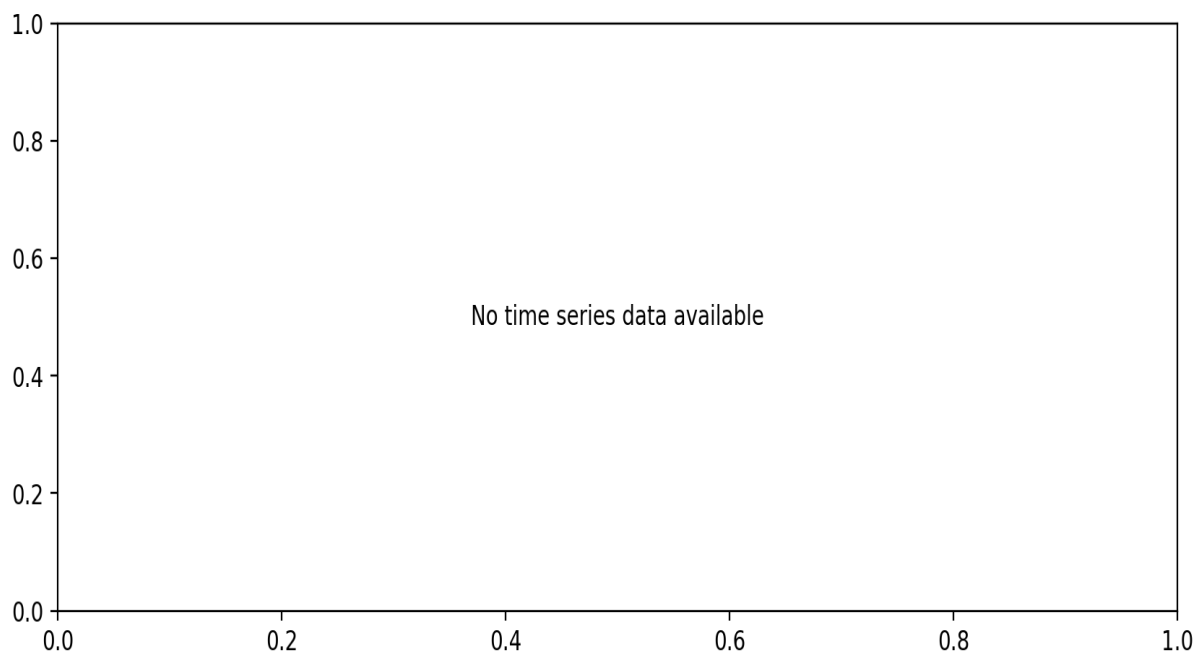


Table: dbo_F_AR_Detail

Rows	520089			
Columns	45			
Column	Type	Non-null	Nulls	Unique
AR Detail Key	int64	520089	0	520089
Company Code	object	520089	0	3
Source Code	object	520089	0	1
Record Type	int64	520089	0	1
AR Detail Id	int64	520089	0	520089
AR Header Key	int64	520089	0	252939
Business Unit Key	int64	520089	0	7
Company Key	int64	520089	0	3
Currency Key	int64	520089	0	3
Customer Key	int64	520089	0	1855

Insights

- Column 'Source Code' in 'dbo_F_AR_Detail' has only one unique value (['Demo']).
- Column 'Record Type' in 'dbo_F_AR_Detail' has only one unique value ([0]).
- Column 'Division Key' in 'dbo_F_AR_Detail' has only one unique value ([-1]).
- Column 'Facility Key' in 'dbo_F_AR_Detail' has only one unique value ([-1]).
- Column 'Document Type' in 'dbo_F_AR_Detail' has only one unique value (['Invoice']).
- Column 'Deleted Flag' in 'dbo_F_AR_Detail' has only one unique value ([0]).
- Column 'Source Log Key' in 'dbo_F_AR_Detail' has only one unique value ([2]).
- Column 'Analytics Log Key' in 'dbo_F_AR_Detail' has only one unique value ([0]).
- Column 'Dimension 01 Key' in 'dbo_F_AR_Detail' has only one unique value ([-1]).
- Column 'Dimension 02 Key' in 'dbo_F_AR_Detail' has only one unique value ([-1]).
- Column 'Dimension 03 Key' in 'dbo_F_AR_Detail' has only one unique value ([-1]).
- Column 'Dimension 04 Key' in 'dbo_F_AR_Detail' has only one unique value ([-1]).
- Column 'Dimension 05 Key' in 'dbo_F_AR_Detail' has only one unique value ([-1]).
- Column 'Dimension 06 Key' in 'dbo_F_AR_Detail' has only one unique value ([-1]).
- Column 'Dimension 07 Key' in 'dbo_F_AR_Detail' has only one unique value ([-1]).
- Column 'Dimension 08 Key' in 'dbo_F_AR_Detail' has only one unique value ([-1]).
- Column 'Dimension 09 Key' in 'dbo_F_AR_Detail' has only one unique value ([-1]).
- Column 'Dimension 10 Key' in 'dbo_F_AR_Detail' has only one unique value ([-1]).
- Numeric column 'Txn Amount' in 'dbo_F_AR_Detail' has 111678 potential outliers (IQR method).
- Numeric column 'Txn Amount (TCY)' in 'dbo_F_AR_Detail' has 111678 potential outliers (IQR method).
- Numeric column 'Debit Amount' in 'dbo_F_AR_Detail' has 79772 potential outliers (IQR method).
- Numeric column 'Credit Amount' in 'dbo_F_AR_Detail' has 78008 potential outliers (IQR method).
- Numeric column 'Age Band Days' in 'dbo_F_AR_Detail' has 91416 potential outliers (IQR method).
- Numeric column 'Age Band Sort' in 'dbo_F_AR_Detail' has 13437 potential outliers (IQR method).
- Strong correlations detected in 'dbo_F_AR_Detail': AR Detail Key-AR Detail Id (0.89), AR Detail Key-AR Header Id (0.89), AR Detail Id-Company Key (0.96), AR Detail Id-Currency Key (0.96), AR Detail Id-AR Header Id (0.99), Company Key-Currency Key (1.00), Company Key-AR Header Id (0.98), Currency Key-AR Header Id (0.98), Customer Key-Customer Category Hrchy Key (0.87), Customer Category Hrchy Key-Sales Organization Key (0.85), Customer Category Hrchy Key-Txn Sales Organization Key (0.85), Posting Date Key-Txn Date Key (1.00), Sales Organization Key-Txn Sales Organization Key (1.00), Txn Amount-Txn Amount (TCY) (1.00). Consider multicollinearity handling.
- Numeric column 'Txn Amount' in 'dbo_F_AR_Detail' is highly skewed (skew=2.03). Consider transformation.
- Numeric column 'Txn Amount (TCY)' in 'dbo_F_AR_Detail' is highly skewed (skew=2.03). Consider transformation.
- Numeric column 'Debit Amount' in 'dbo_F_AR_Detail' is highly skewed (skew=26.42). Consider transformation.
- Numeric column 'Credit Amount' in 'dbo_F_AR_Detail' is highly skewed (skew=24.27). Consider transformation.
- Numeric column 'Age Band Days' in 'dbo_F_AR_Detail' is highly skewed (skew=2.64). Consider transformation.

Chart: chart1.png

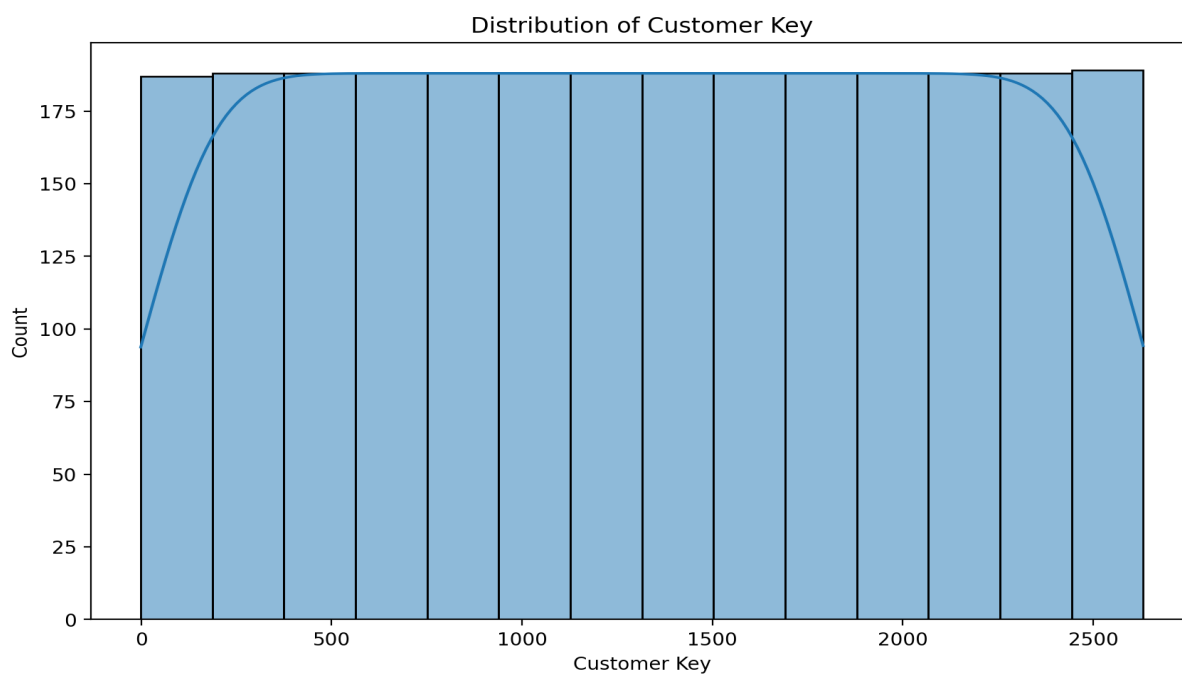


Chart: chart2.png

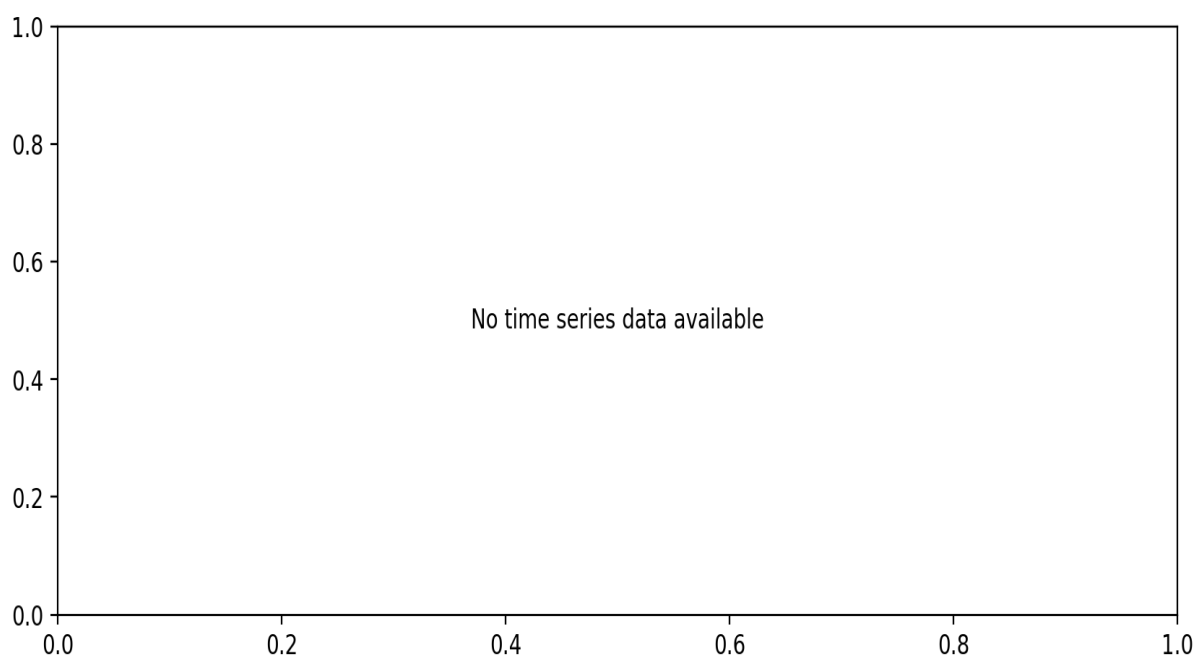


Table: dbo_F_Sales_Goal

Rows	26350			
Columns	32			
Column	Type	Non-null	Nulls	Unique
Sales Goal Key	int64	26350	0	26350
Company Code	object	26350	0	3
Source Code	object	26350	0	1
Dataset	object	26350	0	2
Sales Goal Id	int64	26350	0	25018
Business Unit Key	int64	26350	0	8
Company Key	int64	26350	0	3
Currency Key	int64	26350	0	3
Customer Key	int64	26350	0	1
Customer Category Hrchy Key	int64	26350	0	19

Insights

- In table 'dbo_F_Sales_Goal', columns ['Profit Goal Amount', 'Customer Number', 'Facility Code', 'Sales Goal Quantity', 'Item Number', 'Line Type'] have high missingness (>30%). Consider data collection or imputation.
- Column 'Source Code' in 'dbo_F_Sales_Goal' has only one unique value (['Demo']).
- Column 'Customer Key' in 'dbo_F_Sales_Goal' has only one unique value ([-1]).
- Column 'Customer Geography Hrchy Key' in 'dbo_F_Sales_Goal' has only one unique value ([-1]).
- Column 'Division Key' in 'dbo_F_Sales_Goal' has only one unique value ([-1]).
- Column 'Facility Key' in 'dbo_F_Sales_Goal' has only one unique value ([-1]).
- Column 'Division Code' in 'dbo_F_Sales_Goal' has only one unique value (['WHLS']).
- Column 'Facility Code' in 'dbo_F_Sales_Goal' is entirely empty.
- Column 'Customer Number' in 'dbo_F_Sales_Goal' is entirely empty.
- Column 'Line Type' in 'dbo_F_Sales_Goal' has only one unique value (['Item']).
- Column 'Profit Goal Amount' in 'dbo_F_Sales_Goal' is entirely empty.
- Column 'Deleted Flag' in 'dbo_F_Sales_Goal' has only one unique value ([0]).
- Column 'Source Log Key' in 'dbo_F_Sales_Goal' has only one unique value ([0]).
- Numeric column 'Item Key' in 'dbo_F_Sales_Goal' has 1332 potential outliers (IQR method).
- Numeric column 'Sales Goal Quantity' in 'dbo_F_Sales_Goal' has 64 potential outliers (IQR method).
- Numeric column 'Sales Goal Amount' in 'dbo_F_Sales_Goal' has 1986 potential outliers (IQR method).
- Strong correlations detected in 'dbo_F_Sales_Goal': Sales Goal Id-Salesperson Code (0.87), Business Unit Key-Customer Category Hrchy Key (0.89), Business Unit Key-Sales Organization Key (0.90), Company Key-Currency Key (1.00), Company Key-Salesperson Code (1.00), Currency Key-Salesperson Code (1.00), Customer Category Hrchy Key-Sales Organization Key (0.90), Sales Goal Quantity-Sales Goal Amount (0.87). Consider multicollinearity handling.
- Numeric column 'Item Key' in 'dbo_F_Sales_Goal' is highly skewed (skew=5.35). Consider transformation.
- Numeric column 'Sales Goal Quantity' in 'dbo_F_Sales_Goal' is highly skewed (skew=2.72). Consider transformation.
- Numeric column 'Sales Goal Amount' in 'dbo_F_Sales_Goal' is highly skewed (skew=4.73). Consider transformation.

Chart: chart1.png

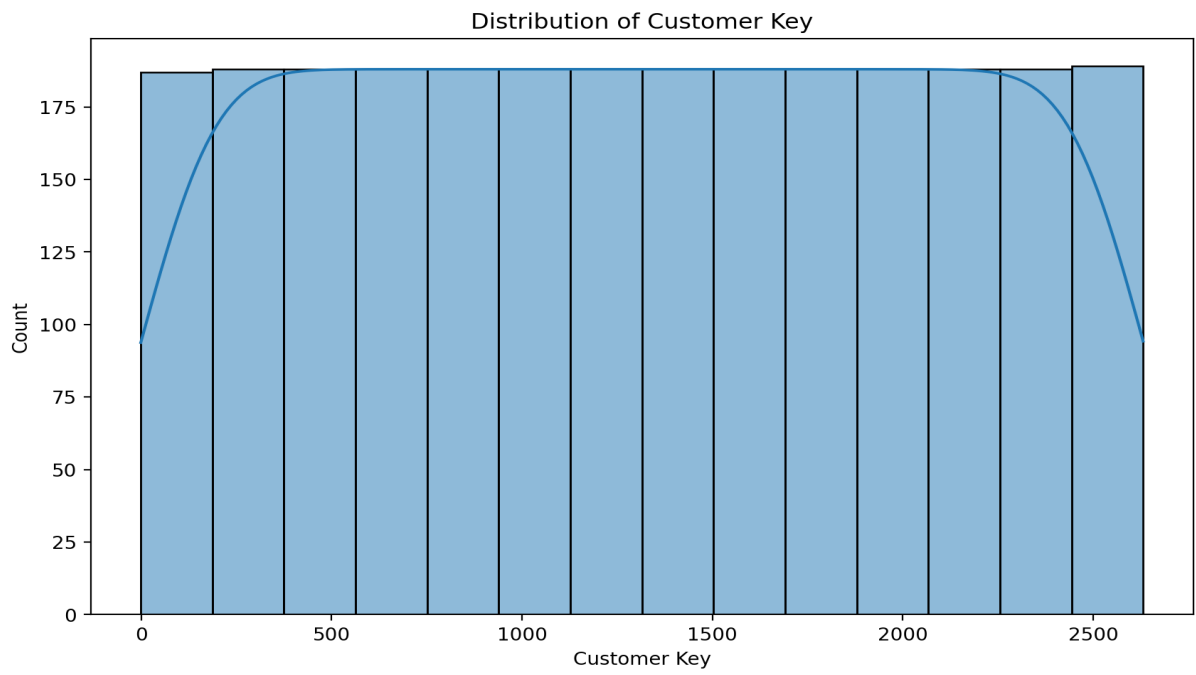


Chart: chart2.png

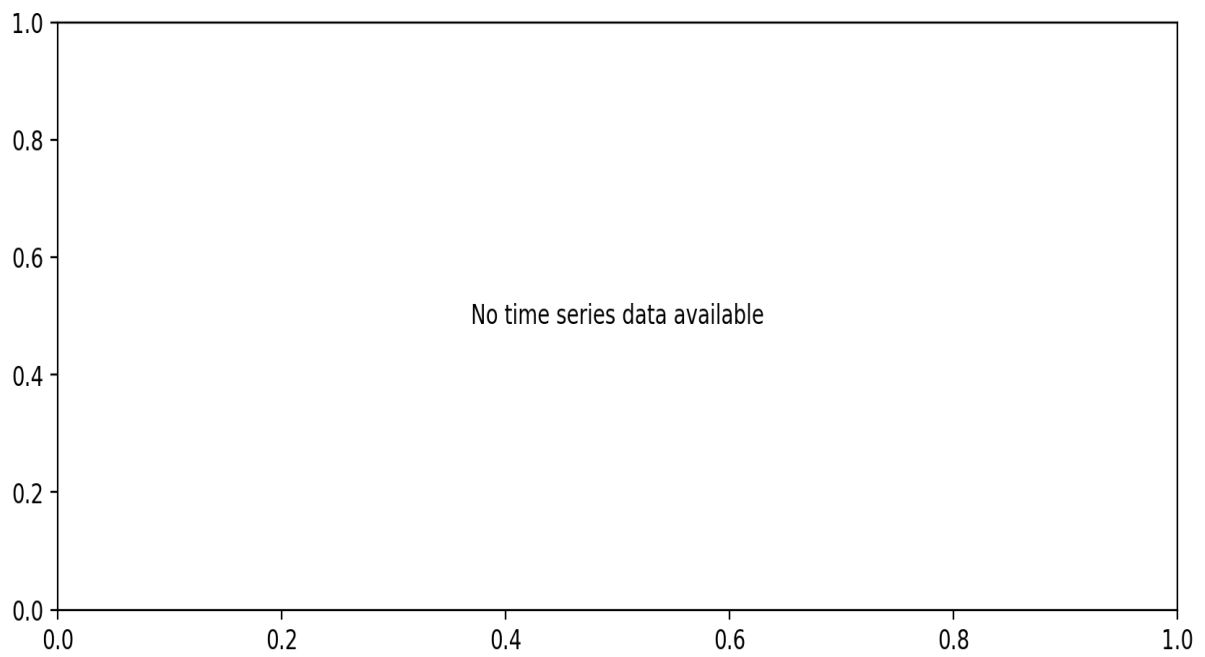


Table: dbo_F_Sales_Order

Rows 80226

Columns 83

Column	Type	Non-null	Nulls	Unique
Sales Order Key	int64	80226	0	80226
Company Code	object	80226	0	3
Source Code	object	80226	0	1
Record Type	int64	80226	0	1
Sales Order Document	object	80226	0	80226
Business Unit Key	int64	80226	0	7
Company Key	int64	80226	0	3
Currency Key	int64	80226	0	3
Customer Key	int64	80226	0	2555
Customer Category Hrchy Key	int64	80226	0	18

Insights

- In table 'dbo_F_Sales_Order', columns ['Product Posting Group', 'Service Item Number', 'Posting Time', 'Warehouse Code', 'Location Code', 'Revision Code', 'Item Variant Code', 'Unit of Measure', 'Requested Ship Date', 'Customer Posting Group', 'Business Posting Group Code', 'Business Posting Group', 'Product Posting Group Code', 'Sales Event Code', 'Discount Reason', 'Discount Reason Code'] have high missingness (>30%). Consider data collection or imputation.
- Column 'Source Code' in 'dbo_F_Sales_Order' has only one unique value (['Demo']).
- Column 'Record Type' in 'dbo_F_Sales_Order' has only one unique value ([0]).
- Column 'Sales Order Document' in 'dbo_F_Sales_Order' appears to be a high-cardinality text field (unique ratio >90%).
- Column 'Facility Key' in 'dbo_F_Sales_Order' has only one unique value ([-1]).
- Column 'Item Variant Key' in 'dbo_F_Sales_Order' has only one unique value ([-1]).
- Column 'Location Key' in 'dbo_F_Sales_Order' has only one unique value ([-1]).
- Column 'Service Item Key' in 'dbo_F_Sales_Order' has only one unique value ([-1]).
- Column 'Warehouse Key' in 'dbo_F_Sales_Order' has only one unique value ([-1]).
- Column 'Sales Order Type' in 'dbo_F_Sales_Order' has only one unique value (['Sales Order']).
- Column 'Posting Time' in 'dbo_F_Sales_Order' is entirely empty.
- Column 'Warehouse Code' in 'dbo_F_Sales_Order' is entirely empty.
- Column 'Location Code' in 'dbo_F_Sales_Order' is entirely empty.
- Column 'Line Type' in 'dbo_F_Sales_Order' has only one unique value (['Item']).
- Column 'Revision Code' in 'dbo_F_Sales_Order' is entirely empty.
- Column 'Item Variant Code' in 'dbo_F_Sales_Order' is entirely empty.
- Column 'Service Item Number' in 'dbo_F_Sales_Order' is entirely empty.
- Column 'Dimension Set Id' in 'dbo_F_Sales_Order' has only one unique value ([0]).
- Column 'Order Status' in 'dbo_F_Sales_Order' has only one unique value (['Open']).
- Column 'Customer Posting Group' in 'dbo_F_Sales_Order' is entirely empty.
- Column 'Business Posting Group Code' in 'dbo_F_Sales_Order' is entirely empty.
- Column 'Business Posting Group' in 'dbo_F_Sales_Order' is entirely empty.
- Column 'Product Posting Group Code' in 'dbo_F_Sales_Order' is entirely empty.
- Column 'Product Posting Group' in 'dbo_F_Sales_Order' is entirely empty.
- Column 'Requested Ship Date' in 'dbo_F_Sales_Order' is entirely empty.
- Column 'Unit of Measure' in 'dbo_F_Sales_Order' is entirely empty.
- Column 'Return Quantity' in 'dbo_F_Sales_Order' has only one unique value ([0.0]).
- Column 'Return Amount' in 'dbo_F_Sales_Order' has only one unique value ([0.0]).
- Column 'Return Amount (TCY)' in 'dbo_F_Sales_Order' has only one unique value ([0.0]).
- Column 'Number Order Lines' in 'dbo_F_Sales_Order' has only one unique value ([1]).
- Column 'Number Return Lines' in 'dbo_F_Sales_Order' has only one unique value ([0]).
- Column 'Deleted Flag' in 'dbo_F_Sales_Order' has only one unique value ([0]).

- Column 'Source Log Key' in 'dbo_F_Sales_Order' has only one unique value ([0]).
- Column 'Analytics Log Key' in 'dbo_F_Sales_Order' has only one unique value ([0]).
- Column 'Dimension 01 Key' in 'dbo_F_Sales_Order' has only one unique value ([1]).
- Column 'Dimension 02 Key' in 'dbo_F_Sales_Order' has only one unique value ([1]).
- Column 'Dimension 03 Key' in 'dbo_F_Sales_Order' has only one unique value ([1]).
- Column 'Dimension 04 Key' in 'dbo_F_Sales_Order' has only one unique value ([1]).
- Column 'Dimension 05 Key' in 'dbo_F_Sales_Order' has only one unique value ([1]).
- Column 'Dimension 06 Key' in 'dbo_F_Sales_Order' has only one unique value ([1]).
- Column 'Dimension 07 Key' in 'dbo_F_Sales_Order' has only one unique value ([1]).
- Column 'Dimension 08 Key' in 'dbo_F_Sales_Order' has only one unique value ([1]).
- Column 'Dimension 09 Key' in 'dbo_F_Sales_Order' has only one unique value ([1]).
- Column 'Dimension 10 Key' in 'dbo_F_Sales_Order' has only one unique value ([1]).
- Numeric column 'Sales Event Key' in 'dbo_F_Sales_Order' has 3353 potential outliers (IQR method).
- Numeric column 'Order Quantity' in 'dbo_F_Sales_Order' has 645 potential outliers (IQR method).
- Numeric column 'Order Amount' in 'dbo_F_Sales_Order' has 4159 potential outliers (IQR method).
- Numeric column 'Order Amount (TCY)' in 'dbo_F_Sales_Order' has 4159 potential outliers (IQR method).
- Numeric column 'Net Order Quantity' in 'dbo_F_Sales_Order' has 645 potential outliers (IQR method).
- Numeric column 'Net Order Amount' in 'dbo_F_Sales_Order' has 4159 potential outliers (IQR method).
- Numeric column 'Net Order Amount (TCY)' in 'dbo_F_Sales_Order' has 4159 potential outliers (IQR method).
- Numeric column 'Cost Amount' in 'dbo_F_Sales_Order' has 5361 potential outliers (IQR method).
- Numeric column 'Gross Profit Amount' in 'dbo_F_Sales_Order' has 3700 potential outliers (IQR method).
- Numeric column 'Discount Amount' in 'dbo_F_Sales_Order' has 12841 potential outliers (IQR method).
- Numeric column 'Discount Amount (TCY)' in 'dbo_F_Sales_Order' has 12841 potential outliers (IQR method).
- Numeric column 'Excluded Flag' in 'dbo_F_Sales_Order' has 8302 potential outliers (IQR method).
- Strong correlations detected in 'dbo_F_Sales_Order': Sales Order Key-Sales Order Header Key (1.00), Business Unit Key-Customer Category Hrchy Key (0.86), Business Unit Key-Item Key (0.86), Business Unit Key-Sales Organization Key (0.87), Business Unit Key-Txn Sales Organization Key (0.87), Company Key-Currency Key (1.00), Company Key-Division Key (1.00), Currency Key-Division Key (1.00), Customer Key-Customer Category Hrchy Key (0.88), Customer Key-Customer Ship-to Key (1.00), Customer Key-Item Key (0.87), Customer Category Hrchy Key-Customer Geography Hrchy Key (0.85), Customer Category Hrchy Key-Customer Ship-to Key (0.88), Customer Category Hrchy Key-Item Key (0.90), Customer Category Hrchy Key-Sales Organization Key (0.87), Customer Category Hrchy Key-Txn Sales Organization Key (0.87), Customer Geography Hrchy Key-Item Key (0.85), Customer Ship-to Key-Item Key (0.87), Item Key-Sales Organization Key (0.87), Item Key-Txn Sales Organization Key (0.87), Order Date Key-Posting Date Key (1.00), Sales Organization Key-Txn Sales Organization Key (1.00), Order Quantity-Net Order Quantity (1.00), Order Amount-Order Amount (TCY) (1.00), Order Amount-Net Order Amount (1.00), Order Amount-Net Order Amount (TCY) (1.00), Order Amount-Cost Amount (0.99), Order Amount-Gross Profit Amount (0.98), Order Amount (TCY)-Net Order Amount (1.00), Order Amount (TCY)-Net Order Amount (TCY) (1.00), Order Amount (TCY)-Cost Amount (0.99), Order Amount (TCY)-Gross Profit Amount (0.98), Net Order Amount-Net Order Amount (TCY) (1.00), Net Order Amount-Cost Amount (0.99), Net Order Amount-Gross Profit Amount (0.98), Net Order Amount (TCY)-Cost Amount (0.99), Net Order Amount (TCY)-Gross Profit Amount (0.98), Cost Amount-Gross Profit Amount (0.94), Discount Amount-Discount Amount (TCY) (1.00). Consider multicollinearity handling.
- Numeric column 'Sales Event Key' in 'dbo_F_Sales_Order' is highly skewed (skew=5.05). Consider transformation.
- Numeric column 'Discount Amount' in 'dbo_F_Sales_Order' is highly skewed (skew=5.88). Consider transformation.
- Numeric column 'Discount Amount (TCY)' in 'dbo_F_Sales_Order' is highly skewed (skew=5.88). Consider transformation.
- Numeric column 'Excluded Flag' in 'dbo_F_Sales_Order' is highly skewed (skew=2.60). Consider transformation.

Chart: chart1.png

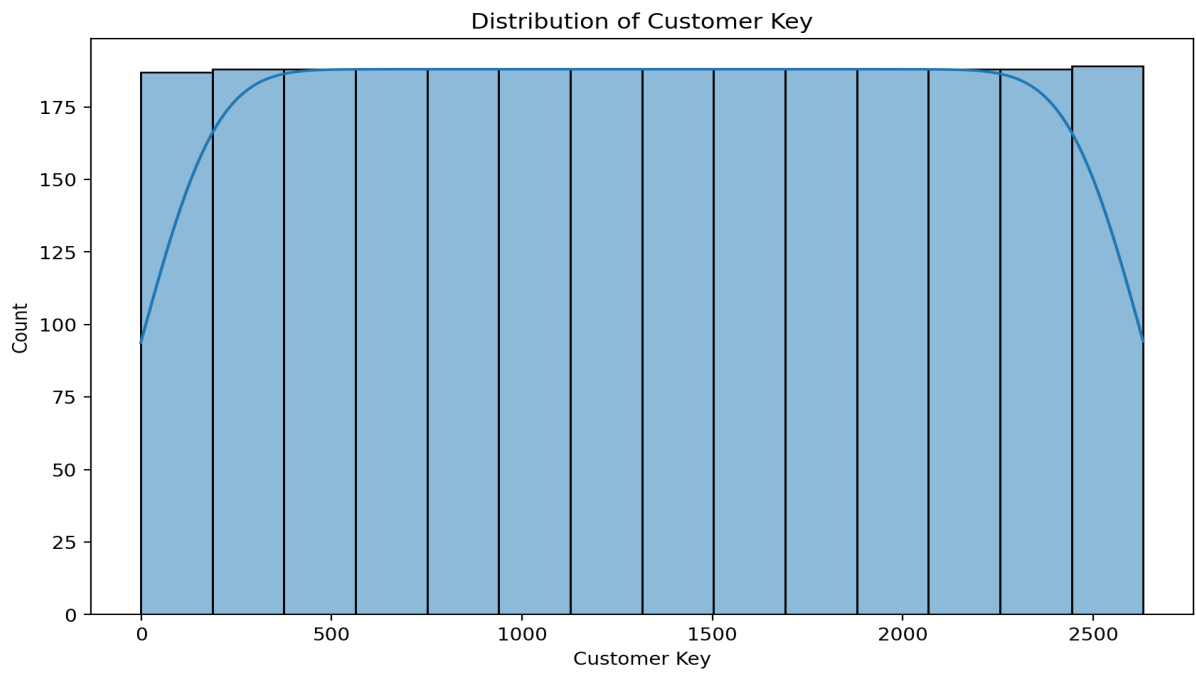


Chart: chart2.png

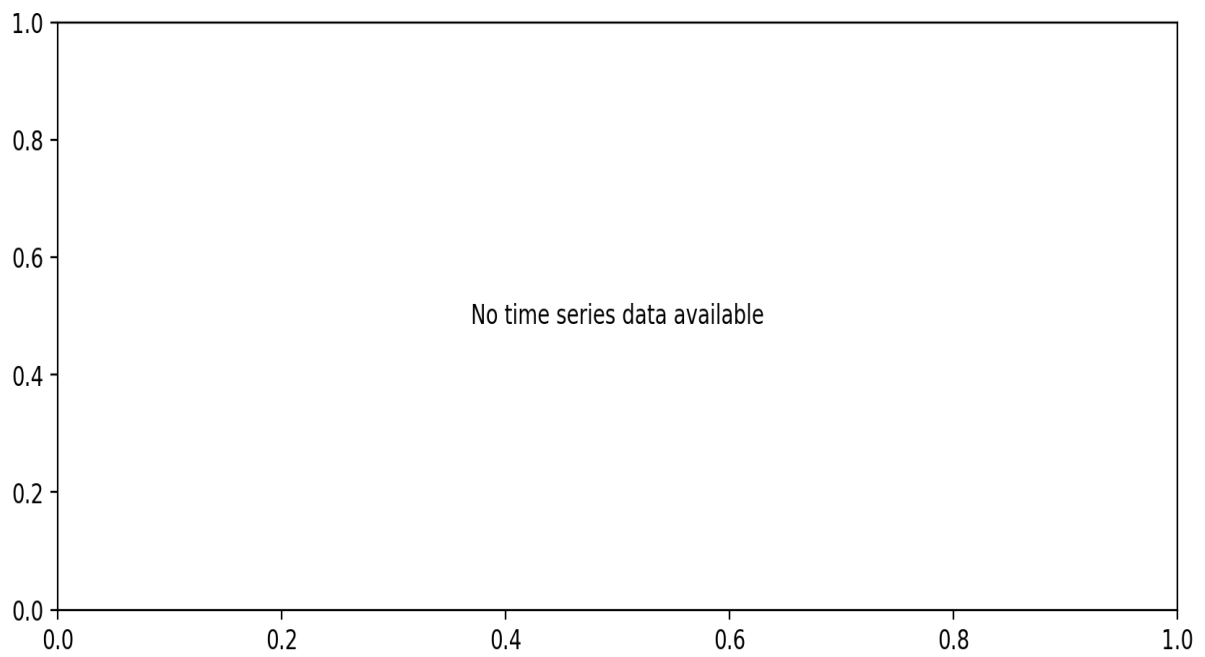


Table: dbo_F_Sales_Pipeline_Snapshot

Rows	31476			
Columns	34			
Column	Type	Non-null	Nulls	Unique
Pipeline Snapshot Key	int64	31476	0	31476
Business Unit Key	int64	31476	0	7
Company Key	int64	31476	0	3
Customer Key	int64	31476	0	1623
Customer Category Hrchy Key	int64	31476	0	18
Customer Geography Hrchy Key	int64	31476	0	98
Division Key	int64	31476	0	3
Facility Key	int64	31476	0	1
Sales Organization Key	int64	31476	0	20
Snapshot Date Key	int64	31476	0	51

Insights

- In table 'dbo_F_Sales_Pipeline_Snapshot', columns ['Analytics Log Key', 'Expected Delivery Band', 'Class Band Sort', 'Expected Delivery Band Sort'] have high missingness (>30%). Consider data collection or imputation.
- Column 'Facility Key' in 'dbo_F_Sales_Pipeline_Snapshot' has only one unique value ([-1]).
- Column 'Warehouse Key' in 'dbo_F_Sales_Pipeline_Snapshot' has only one unique value ([-1]).
- Column 'Class Band Sort' in 'dbo_F_Sales_Pipeline_Snapshot' is entirely empty.
- Column 'Expected Delivery Band' in 'dbo_F_Sales_Pipeline_Snapshot' is entirely empty.
- Column 'Expected Delivery Band Sort' in 'dbo_F_Sales_Pipeline_Snapshot' is entirely empty.
- Column 'Deleted Flag' in 'dbo_F_Sales_Pipeline_Snapshot' has only one unique value ([0]).
- Column 'Analytics Log Key' in 'dbo_F_Sales_Pipeline_Snapshot' is entirely empty.
- Numeric column 'Quote Quantity' in 'dbo_F_Sales_Pipeline_Snapshot' has 172 potential outliers (IQR method).
- Numeric column 'Quote Amount' in 'dbo_F_Sales_Pipeline_Snapshot' has 1052 potential outliers (IQR method).
- Numeric column 'Gross Profit Amount' in 'dbo_F_Sales_Pipeline_Snapshot' has 708 potential outliers (IQR method).
- Numeric column 'Discount Amount' in 'dbo_F_Sales_Pipeline_Snapshot' has 6484 potential outliers (IQR method).
- Numeric column 'Number Quote Lines' in 'dbo_F_Sales_Pipeline_Snapshot' has 682 potential outliers (IQR method).
- Strong correlations detected in 'dbo_F_Sales_Pipeline_Snapshot': Business Unit Key-Customer Category Hrchy Key (0.87), Business Unit Key-Sales Organization Key (0.88), Company Key-Division Key (1.00), Customer Key-Customer Category Hrchy Key (0.88), Customer Category Hrchy Key-Customer Geography Hrchy Key (0.85), Customer Category Hrchy Key-Sales Organization Key (0.86), Quote Amount-Gross Profit Amount (0.97). Consider multicollinearity handling.
- Numeric column 'Discount Amount' in 'dbo_F_Sales_Pipeline_Snapshot' is highly skewed (skew=5.02). Consider transformation.
- Numeric column 'Number Quote Lines' in 'dbo_F_Sales_Pipeline_Snapshot' is highly skewed (skew=6.57). Consider transformation.

Chart: chart1.png

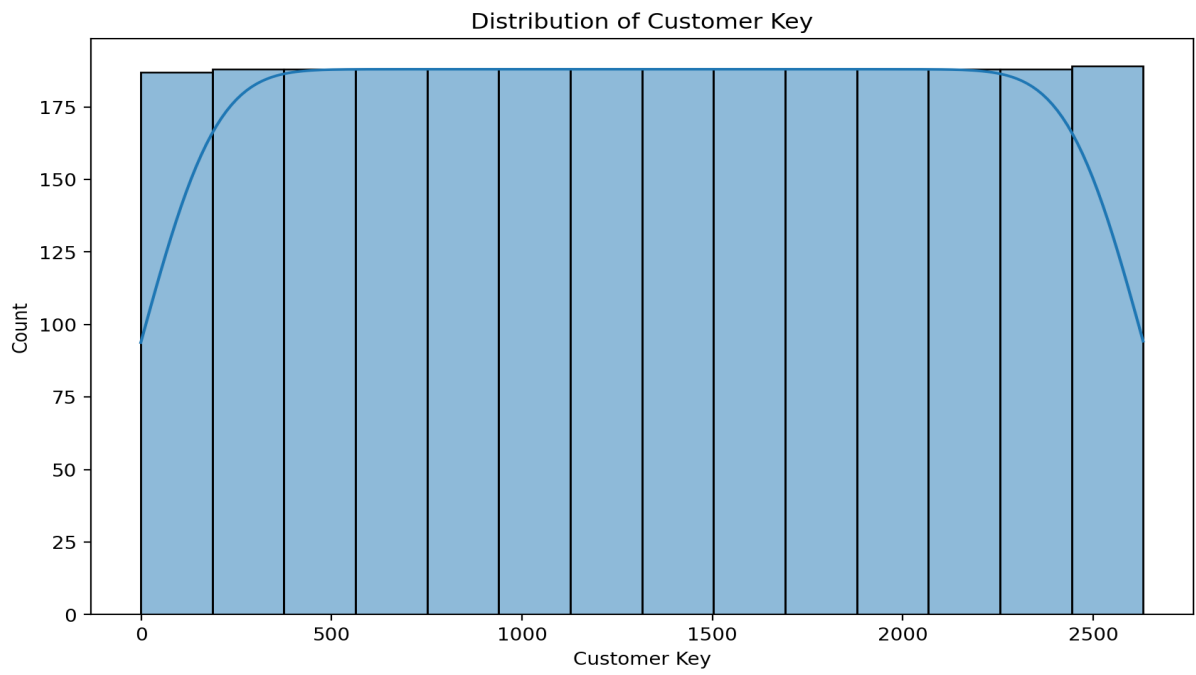


Chart: chart2.png

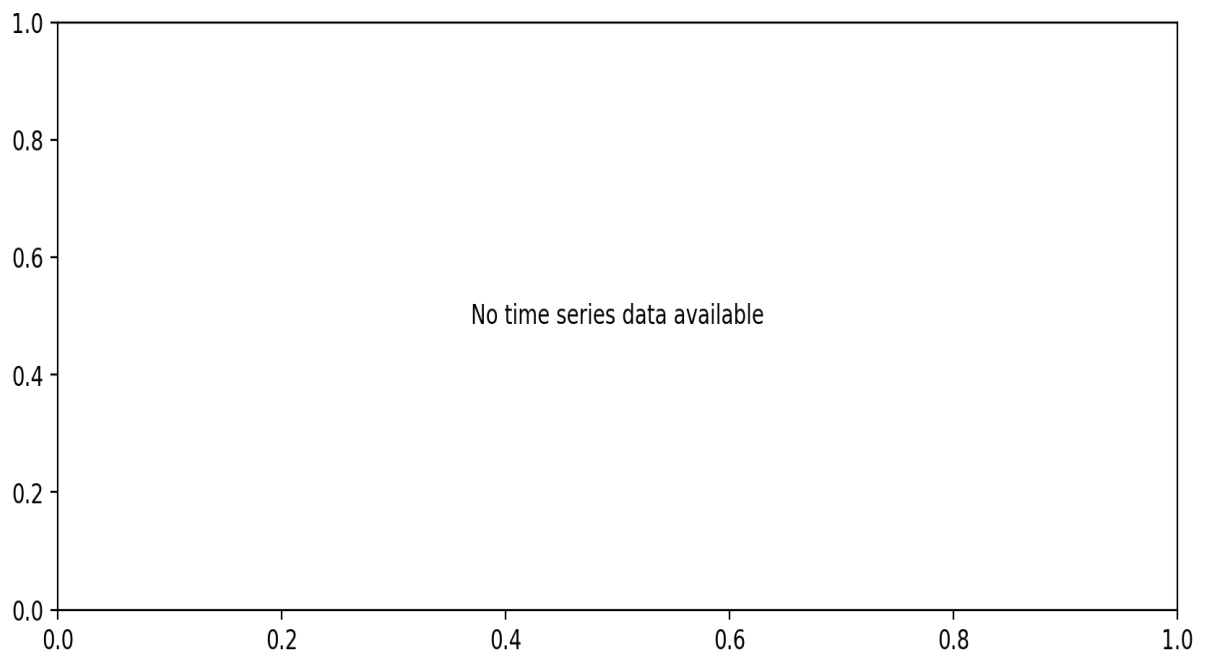


Table: dbo_F_Sales_Transaction

Rows	81423			
Columns	101			
Column	Type	Non-null	Nulls	Unique
Sales Txn Key	int64	81423	0	81423
Company Code	object	81423	0	3
Source Code	object	81423	0	1
Record Type	int64	81423	0	1
Sales Txn Document	object	81423	0	81423
Business Unit Key	int64	81423	0	7
Company Key	int64	81423	0	3
Currency Key	int64	81423	0	3
Customer Key	int64	81423	0	2547
Customer Category Hrchy Key	int64	81423	0	18

Insights

- In table 'dbo_F_Sales_Transaction', columns ['Tax 2 Code', 'Return Reason Code', 'Return Order Document', 'Unit of Measure', 'Service Item Number', 'Item Variant Code', 'Revision Code', 'Product Posting Group', 'Product Posting Group Code', 'Business Posting Group', 'Business Posting Group Code', 'Location Code', 'Warehouse Code', 'Posting Time', 'Other Charges Amount (TCY)', 'Tax 1 Code', 'Return Reason', 'Sales Event Code', 'Discount Reason', 'Discount Reason Code'] have high missingness (>30%). Consider data collection or imputation.
- Column 'Source Code' in 'dbo_F_Sales_Transaction' has only one unique value (['Demo']).
- Column 'Record Type' in 'dbo_F_Sales_Transaction' has only one unique value ([0]).
- Column 'Sales Txn Document' in 'dbo_F_Sales_Transaction' appears to be a high-cardinality text field (unique ratio >90%).
- Column 'Facility Key' in 'dbo_F_Sales_Transaction' has only one unique value ([-1]).
- Column 'Item Variant Key' in 'dbo_F_Sales_Transaction' has only one unique value ([-1]).
- Column 'Location Key' in 'dbo_F_Sales_Transaction' has only one unique value ([-1]).
- Column 'Service Item Key' in 'dbo_F_Sales_Transaction' has only one unique value ([-1]).
- Column 'Warehouse Key' in 'dbo_F_Sales_Transaction' has only one unique value ([-1]).
- Column 'Posting Time' in 'dbo_F_Sales_Transaction' is entirely empty.
- Column 'Warehouse Code' in 'dbo_F_Sales_Transaction' is entirely empty.
- Column 'Location Code' in 'dbo_F_Sales_Transaction' is entirely empty.
- Column 'Business Posting Group Code' in 'dbo_F_Sales_Transaction' is entirely empty.
- Column 'Business Posting Group' in 'dbo_F_Sales_Transaction' is entirely empty.
- Column 'Product Posting Group Code' in 'dbo_F_Sales_Transaction' is entirely empty.
- Column 'Product Posting Group' in 'dbo_F_Sales_Transaction' is entirely empty.
- Column 'Line Type' in 'dbo_F_Sales_Transaction' has only one unique value (['Item']).
- Column 'Revision Code' in 'dbo_F_Sales_Transaction' is entirely empty.
- Column 'Item Variant Code' in 'dbo_F_Sales_Transaction' is entirely empty.
- Column 'Service Item Number' in 'dbo_F_Sales_Transaction' is entirely empty.
- Column 'Dimension Set Id' in 'dbo_F_Sales_Transaction' has only one unique value ([0]).
- Column 'Unit of Measure' in 'dbo_F_Sales_Transaction' is entirely empty.
- Column 'Credit Quantity' in 'dbo_F_Sales_Transaction' has only one unique value ([0.0]).
- Column 'Credit Amount' in 'dbo_F_Sales_Transaction' has only one unique value ([0.0]).
- Column 'Credit Amount (TCY)' in 'dbo_F_Sales_Transaction' has only one unique value ([0.0]).
- Column 'Return Reason Code' in 'dbo_F_Sales_Transaction' is entirely empty.
- Column 'Return Order Document' in 'dbo_F_Sales_Transaction' is entirely empty.
- Column 'Other Charges Amount' in 'dbo_F_Sales_Transaction' has only one unique value ([0.0]).
- Column 'Other Charges Amount (TCY)' in 'dbo_F_Sales_Transaction' is entirely empty.
- Column 'Tax 1 Code' in 'dbo_F_Sales_Transaction' is entirely empty.
- Column 'Tax 1 Amount' in 'dbo_F_Sales_Transaction' has only one unique value ([0.0]).
- Column 'Tax 2 Code' in 'dbo_F_Sales_Transaction' is entirely empty.

- Column 'Tax 2 Amount' in 'dbo_F_Sales_Transaction' has only one unique value ([0.0]).
- Column 'Number Credit Lines' in 'dbo_F_Sales_Transaction' has only one unique value ([0]).
- Column 'Deleted Flag' in 'dbo_F_Sales_Transaction' has only one unique value ([0]).
- Column 'Source Log Key' in 'dbo_F_Sales_Transaction' has only one unique value ([0]).
- Column 'Dimension 01 Key' in 'dbo_F_Sales_Transaction' has only one unique value ([-1]).
- Column 'Dimension 02 Key' in 'dbo_F_Sales_Transaction' has only one unique value ([-1]).
- Column 'Dimension 03 Key' in 'dbo_F_Sales_Transaction' has only one unique value ([-1]).
- Column 'Dimension 04 Key' in 'dbo_F_Sales_Transaction' has only one unique value ([-1]).
- Column 'Dimension 05 Key' in 'dbo_F_Sales_Transaction' has only one unique value ([-1]).
- Column 'Dimension 06 Key' in 'dbo_F_Sales_Transaction' has only one unique value ([-1]).
- Column 'Dimension 07 Key' in 'dbo_F_Sales_Transaction' has only one unique value ([-1]).
- Column 'Dimension 08 Key' in 'dbo_F_Sales_Transaction' has only one unique value ([-1]).
- Column 'Dimension 09 Key' in 'dbo_F_Sales_Transaction' has only one unique value ([-1]).
- Column 'Dimension 10 Key' in 'dbo_F_Sales_Transaction' has only one unique value ([-1]).
- Numeric column 'Sales Event Key' in 'dbo_F_Sales_Transaction' has 3386 potential outliers (IQR method).
- Numeric column 'Sales Quantity' in 'dbo_F_Sales_Transaction' has 631 potential outliers (IQR method).
- Numeric column 'Sales Amount' in 'dbo_F_Sales_Transaction' has 4096 potential outliers (IQR method).
- Numeric column 'Sales Amount (TCY)' in 'dbo_F_Sales_Transaction' has 4096 potential outliers (IQR method).
- Numeric column 'Return Quantity' in 'dbo_F_Sales_Transaction' has 2221 potential outliers (IQR method).
- Numeric column 'Return Amount' in 'dbo_F_Sales_Transaction' has 2221 potential outliers (IQR method).
- Numeric column 'Return Amount (TCY)' in 'dbo_F_Sales_Transaction' has 2221 potential outliers (IQR method).
- Numeric column 'Net Sales Quantity' in 'dbo_F_Sales_Transaction' has 2532 potential outliers (IQR method).
- Numeric column 'Net Sales Amount' in 'dbo_F_Sales_Transaction' has 4711 potential outliers (IQR method).
- Numeric column 'Net Sales Amount (TCY)' in 'dbo_F_Sales_Transaction' has 4711 potential outliers (IQR method).
- Numeric column 'Cost Amount' in 'dbo_F_Sales_Transaction' has 5261 potential outliers (IQR method).
- Numeric column 'Gross Profit Amount' in 'dbo_F_Sales_Transaction' has 4752 potential outliers (IQR method).
- Numeric column 'Discount Amount' in 'dbo_F_Sales_Transaction' has 13935 potential outliers (IQR method).
- Numeric column 'Discount Amount (TCY)' in 'dbo_F_Sales_Transaction' has 13935 potential outliers (IQR method).
- Numeric column 'Number Sales Lines' in 'dbo_F_Sales_Transaction' has 2221 potential outliers (IQR method).
- Numeric column 'Number Return Lines' in 'dbo_F_Sales_Transaction' has 2221 potential outliers (IQR method).
- Numeric column 'Excluded Flag' in 'dbo_F_Sales_Transaction' has 8469 potential outliers (IQR method).
- Numeric column 'gpb Sales Amount' in 'dbo_F_Sales_Transaction' has 5730 potential outliers (IQR method).
- Numeric column 'gpb Return Amount' in 'dbo_F_Sales_Transaction' has 2221 potential outliers (IQR method).
- Numeric column 'gpb Net Sales Amount' in 'dbo_F_Sales_Transaction' has 6435 potential outliers (IQR method).
- Numeric column 'gpb Cost Amount' in 'dbo_F_Sales_Transaction' has 6036 potential outliers (IQR method).
- Numeric column 'gpb Gross Profit Amount' in 'dbo_F_Sales_Transaction' has 5190 potential outliers (IQR method).
- Numeric column 'gpb Discount Amount' in 'dbo_F_Sales_Transaction' has 13541 potential outliers (IQR method).
- Numeric column 'gpb Discount%' in 'dbo_F_Sales_Transaction' has 13452 potential outliers (IQR method).
- Numeric column 'gpb Sales Quantity' in 'dbo_F_Sales_Transaction' has 957 potential outliers (IQR method).
- Numeric column 'gpb Net Sales Quantity' in 'dbo_F_Sales_Transaction' has 3178 potential outliers (IQR method).
- Strong correlations detected in 'dbo_F_Sales_Transaction': Sales Txn Key-Sales Txn Header Key (1.00), Business Unit Key-Customer Category Hrchy Key (0.86), Business Unit Key-Item Key (0.86), Business Unit Key-Sales Organization Key (0.87), Business Unit Key-Txn Sales Organization Key (0.87), Company Key-Currency Key (1.00), Company Key-Division Key (1.00), Currency Key-Division Key (1.00), Customer Key-Customer Category Hrchy Key (0.88), Customer Key-Customer Ship-to Key (1.00), Customer Key-Item Key (0.87), Customer Category Hrchy Key-Customer Geography Hrchy Key (0.85), Customer Category Hrchy Key-Customer Ship-to Key (0.88), Customer Category Hrchy Key-Item Key (0.90), Customer Category Hrchy Key-Sales Organization Key (0.87), Customer Category Hrchy Key-Txn Sales Organization Key (0.87), Customer Geography Hrchy Key-Item Key (0.86), Customer Ship-to Key-Item Key (0.87), Item Key-Sales Organization Key (0.87), Item Key-Txn Sales Organization Key (0.87), Posting Date Key-Txn Date Key (1.00), Sales Organization Key-Txn Sales Organization Key (1.00), Sales Quantity-Net Sales Quantity (0.95), Sales Amount-Sales Amount (TCY) (1.00), Sales Amount-Net Sales Amount (0.97), Sales Amount-Net Sales Amount (TCY) (0.97), Sales Amount-Cost Amount (0.99), Sales Amount-Gross Profit Amount (0.88), Sales Amount-gpb Sales Amount (0.88), Sales Amount-gpb Net Sales Amount (0.86), Sales Amount (TCY)-Net Sales Amount (0.97), Sales Amount (TCY)-Net Sales Amount (TCY) (0.97), Sales Amount (TCY)-Cost Amount (0.99), Sales Amount (TCY)-Gross Profit Amount (0.88), Sales Amount (TCY)-gpb Sales Amount (0.88), Sales Amount (TCY)-gpb Net Sales Amount (0.86), Return Quantity-Number Sales Lines (0.89), Return Quantity-Number Return Lines (0.89), Return Amount-Return Amount (TCY) (1.00), Return Amount-gpb Return Amount (1.00), Return Amount (TCY)-gpb Return Amount (1.00), Net Sales Amount-Net Sales Amount (TCY) (1.00), Net Sales Amount-Cost Amount (0.96), Net Sales Amount-Gross Profit Amount (0.96), Net Sales Amount-gpb Sales Amount (0.85), Net Sales Amount-gpb Net Sales Amount (0.89), Net Sales Amount-gpb

Cost Amount (0.87), Net Sales Amount-gpb Gross Profit Amount (0.87), Net Sales Amount (TCY)-Cost Amount (0.96), Net Sales Amount (TCY)-Gross Profit Amount (0.96), Net Sales Amount (TCY)-gpb Sales Amount (0.85), Net Sales Amount (TCY)-gpb Net Sales Amount (0.89), Net Sales Amount (TCY)-gpb Cost Amount (0.87), Net Sales Amount (TCY)-gpb Gross Profit Amount (0.87), Cost Amount-gpb Sales Amount (0.87), Cost Amount-gpb Net Sales Amount (0.86), Cost Amount-gpb Cost Amount (0.87), Gross Profit Amount-gpb Gross Profit Amount (0.86), Discount Amount-Discount Amount (TCY) (1.00), Discount Amount-gpb Discount Amount (0.92), Discount Amount (TCY)-gpb Discount Amount (0.92), Number Sales Lines-Number Return Lines (1.00), gpb Sales Amount-gpb Net Sales Amount (0.98), gpb Sales Amount-gpb Cost Amount (0.97), gpb Sales Amount-gpb Gross Profit Amount (0.96), gpb Net Sales Amount-gpb Cost Amount (0.98), gpb Net Sales Amount-gpb Gross Profit Amount (0.98), gpb Cost Amount-gpb Gross Profit Amount (0.93), gpb Sales Quantity-gpb Net Sales Quantity (0.96). Consider multicollinearity handling.

- Numeric column 'Sales Event Key' in 'dbo_F_Sales_Transaction' is highly skewed (skew=5.08). Consider transformation.
- Numeric column 'Return Quantity' in 'dbo_F_Sales_Transaction' is highly skewed (skew=7.66). Consider transformation.
- Numeric column 'Return Amount' in 'dbo_F_Sales_Transaction' is highly skewed (skew=11.65). Consider transformation.
- Numeric column 'Return Amount (TCY)' in 'dbo_F_Sales_Transaction' is highly skewed (skew=11.65). Consider transformation.
- Numeric column 'Discount Amount' in 'dbo_F_Sales_Transaction' is highly skewed (skew=5.96). Consider transformation.
- Numeric column 'Discount Amount (TCY)' in 'dbo_F_Sales_Transaction' is highly skewed (skew=5.96). Consider transformation.
- Numeric column 'Number Sales Lines' in 'dbo_F_Sales_Transaction' is highly skewed (skew=-5.80). Consider transformation.
- Numeric column 'Number Return Lines' in 'dbo_F_Sales_Transaction' is highly skewed (skew=5.80). Consider transformation.
- Numeric column 'Excluded Flag' in 'dbo_F_Sales_Transaction' is highly skewed (skew=2.59). Consider transformation.
- Numeric column 'gpb Sales Amount' in 'dbo_F_Sales_Transaction' is highly skewed (skew=2.32). Consider transformation.
- Numeric column 'gpb Return Amount' in 'dbo_F_Sales_Transaction' is highly skewed (skew=11.82). Consider transformation.
- Numeric column 'gpb Discount Amount' in 'dbo_F_Sales_Transaction' is highly skewed (skew=6.73). Consider transformation.
- Numeric column 'gpb Discount%' in 'dbo_F_Sales_Transaction' is highly skewed (skew=2.61). Consider transformation.

Chart: chart1.png

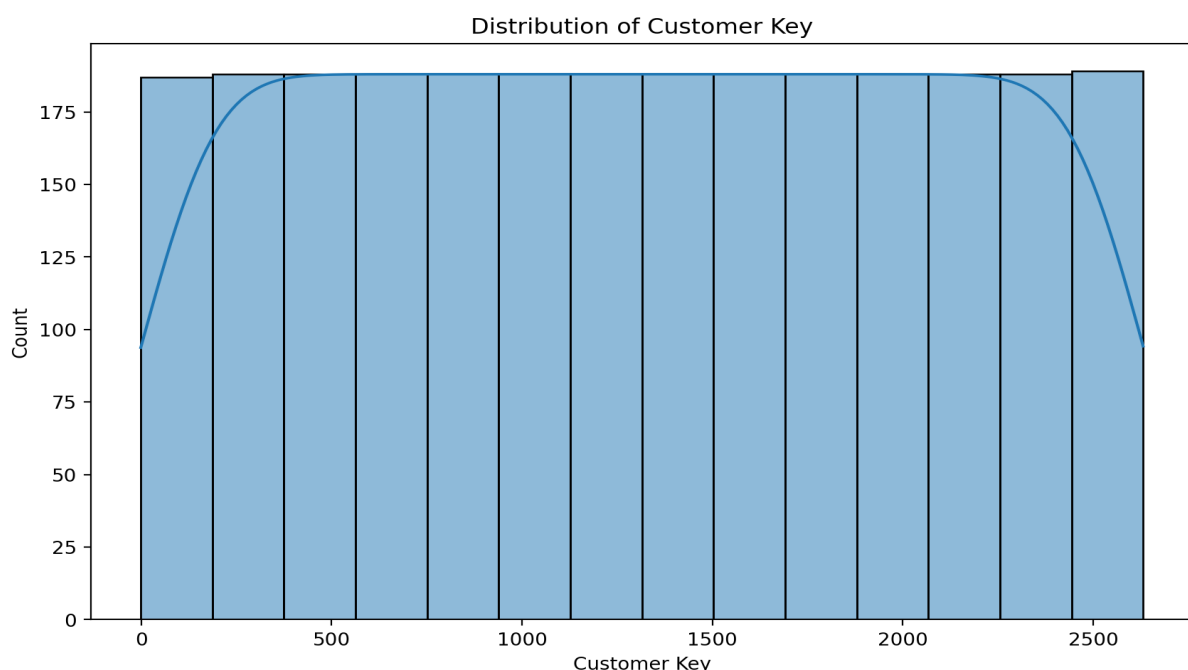
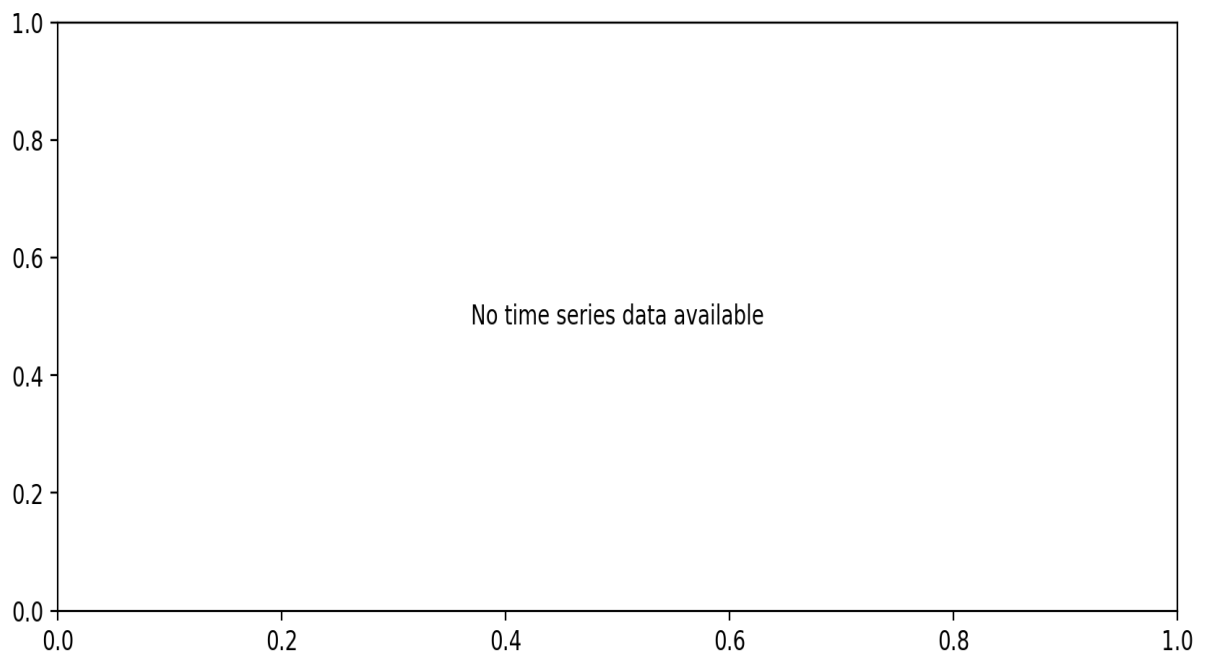


Chart: chart2.png



Appendix - Charts

chart1.png

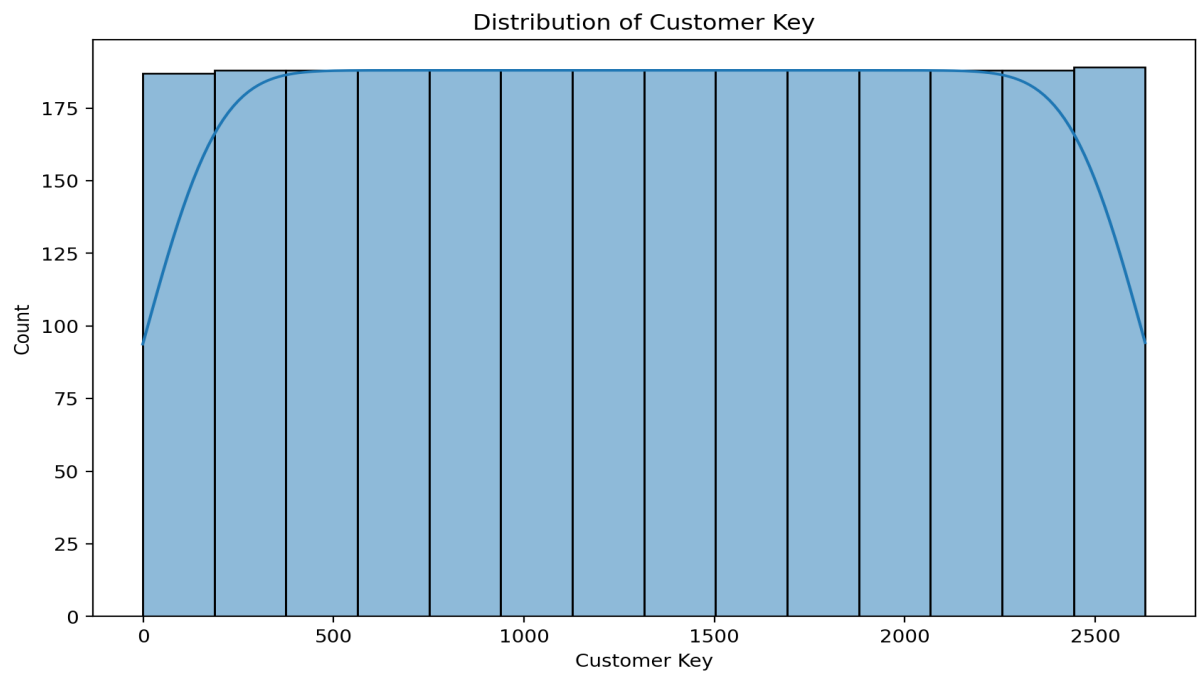


chart2.png

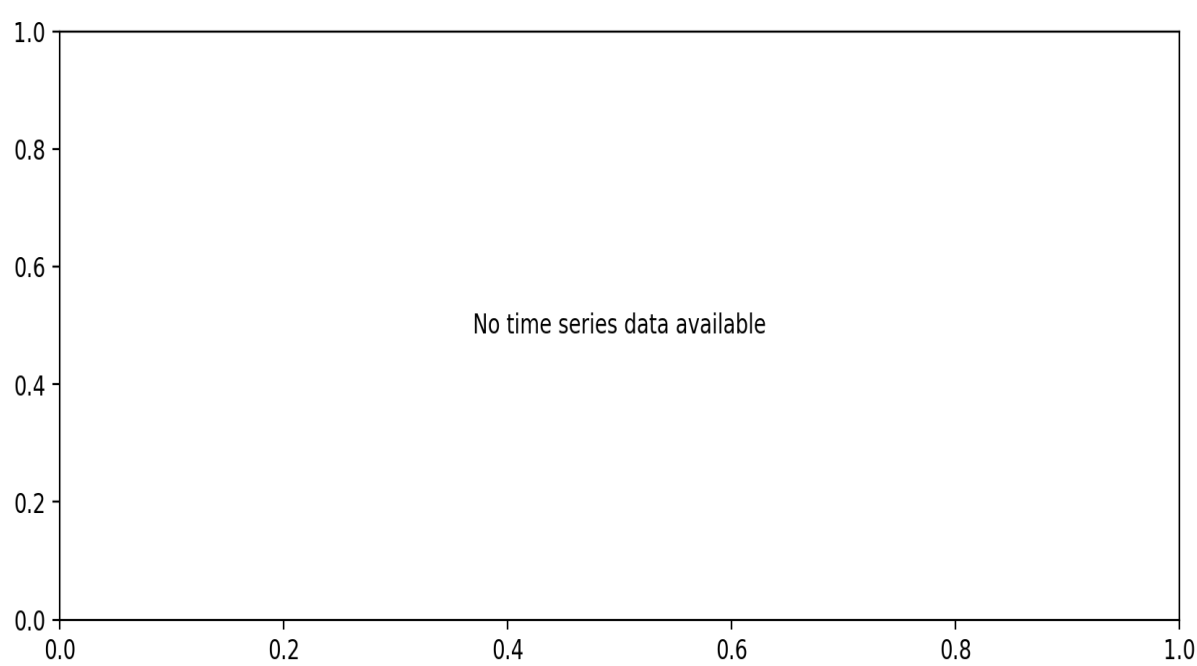


chart3.png

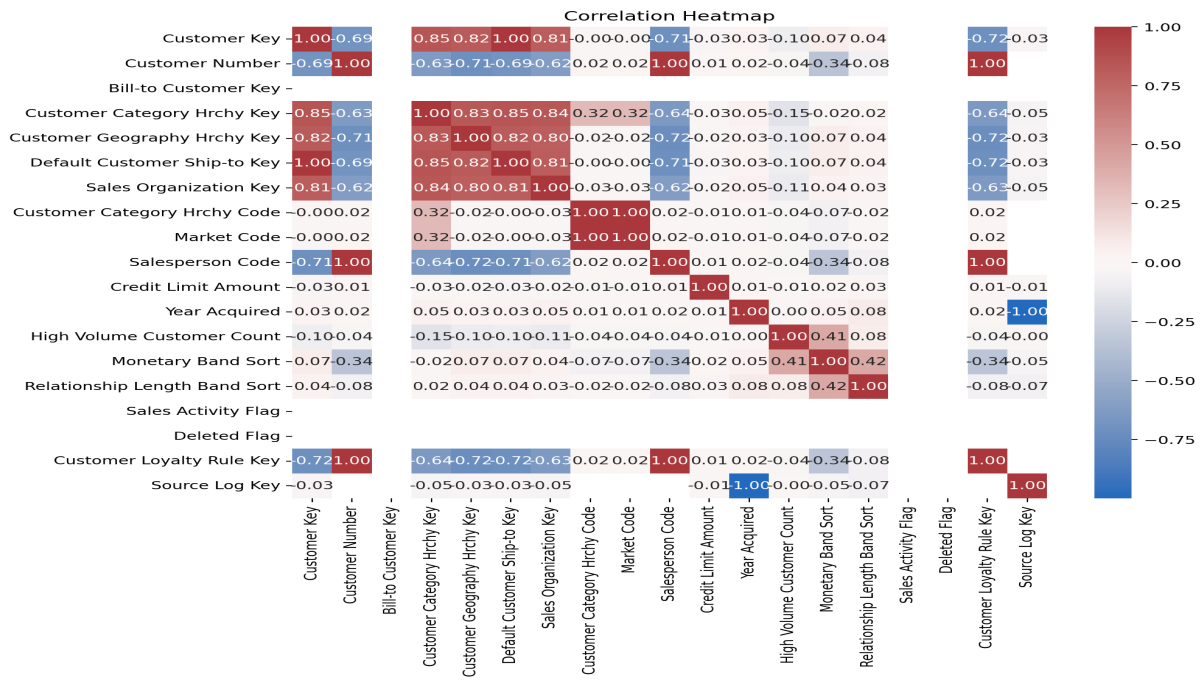


chart4.png

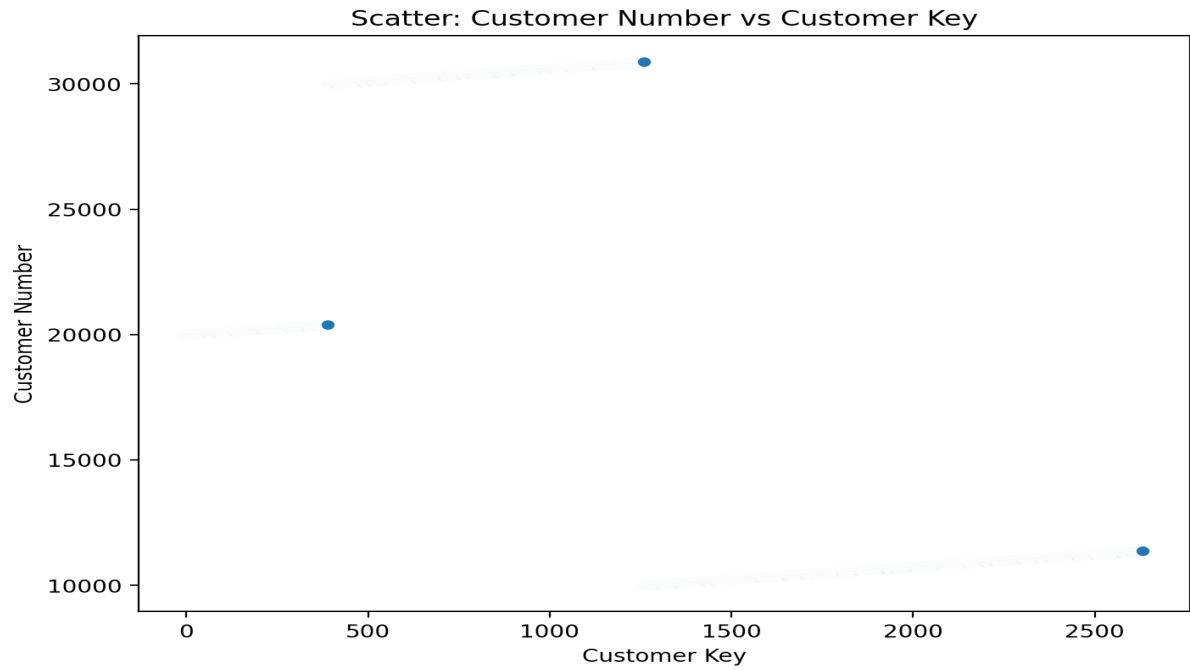


chart5.png

Boxplot of Customer Key

